

Algebra Midterm 2 — Homework and Detailed Proofs

Algebra Midterm 2 — Scope and study map

Homework-aligned study based on the supplied textbook and Weeks 4–6 lecture notes.

This second Algebra study follows the supplied list exactly: **39 Dummit–Foote exercises plus one additional problem**, with separate detailed solutions. It covers §§2.5, 3.1–3.5, 4.1–4.4, 4.6, 5.1, and 5.4. The selected subparts are **3.2.9(a)–(f)**, **4.4.8(b)–(c)**, and **4.4.20(a)–(b)**; all parts are included for the other listed exercises.

The proofs use the lecture methods wherever applicable and the assigned textbook for additional details. Routine steps are explained, but the arguments avoid unnecessary machinery. Solutions and theorem proofs were independently reconstructed rather than copied from an answer key.

Conventions: D_{2n} has order $2n$, permutation products act right to left, and $[x, y] = x^{-1}y^{-1}xy$ follows the book. C_m denotes the cyclic group of order m (the book's Z_m). Setwise and pointwise stabilizers remain distinct.

Sources: the supplied Dummit–Foote book PDF, with printed-page and PDF-page locations in each exercise header; the supplied Weeks 4–6 lecture PDFs, all 36 pages visually inspected; and the additional problem supplied directly in the request. Mathematical proofs here are internal derivations grounded in those definitions and theorems. The lecture map distinguishes context from assigned scope. Exercise 4.1.7(d) has a singleton boundary exception, explained in its solution.

Using the study: read the definitions and theorem statements, attempt each problem without opening its solution, then compare the complete argument with the separate worked solution. The solutions include technique notes and links to their foundations. Check hypotheses before reaching for a theorem: normality, finiteness, commutativity, and surjectivity each do specific work.

All entries begin with review disabled individually. The new folders remain enabled, so you can turn on selected entries later. Earlier Algebra content and review history are preserved.

Definitions

- Normal subgroups, quotient groups, and coset order
- Subgroup lattices and Hasse diagrams
- Centralizers, normalizers, and the center
- Commutators, derived subgroups, and perfectness
- Simple groups and composition series
- Solvable groups
- Actions, orbits, stabilizers, and kernels
- Blocks and primitive permutation actions

- Automorphisms, inner automorphisms, and characteristic subgroups
- p -subgroups, Sylow subgroups, and Hall subgroups
- Direct products, complements, and extremal subgroups
- Sign and alternating groups
- Conjugacy classes and cycle type

Important theorems and detailed proofs

- Cosets, Lagrange, and multiplicativity of index
- Products of subgroups and their orders
- Orbit–stabilizer and the kernel of an action
- First Isomorphism Theorem
- Second (diamond) Isomorphism Theorem
- Third Isomorphism Theorem
- Subgroup correspondence (Lattice Isomorphism Theorem)
- Commutator criterion for abelian quotients
- Derived sequence criterion and solvable extensions
- Conjugacy classes, the class equation, and p -group centers
- Coset actions, core, Cayley, and smallest-prime index
- Inner automorphisms and characteristic-to-normal transfer
- Recognizing an internal direct product
- Existence and uniqueness of composition factors
- Simplicity of A_n for $n \geq 5$
- Orbit counting by fixed points
- Cauchy’s theorem

Proof techniques

- Technique — Define a map before manipulating quotients
- Technique — Prove invariance on generators, then on words
- Technique — Two divisibility constraints on one subgroup or image
- Technique — Choose the set, identify the stabilizer, then count
- Technique — Conjugate supports and pair transpositions
- Technique — Build below and above a normal subgroup
- Technique — Minimality and maximality turn containment into a dichotomy
- Technique — Exhaust maximal subgroups before drawing a lattice

Lecture context

- Lecture map and scope boundaries

Section 2.5 — 3 problems

- 2.5.12 — Subgroup lattice of $C_2 \times C_4$
- 2.5.13 — Subgroup lattice of $C_2 \times C_8$
- 2.5.14 — Same lattice, different groups

Section 3.1 — 5 problems

- 3.1.11 — Upper triangular homomorphisms and fibers
- 3.1.22 — Arbitrary intersections of normal subgroups
- 3.1.28 — Checking a normalizer on generators
- 3.1.36 — A cyclic central quotient forces commutativity
- 3.1.41 — The commutator subgroup and abelianization

Section 3.2 — 5 problems

- 3.2.4 — Center of a group of order pq
- 3.2.8 — Coprime subgroup orders give trivial intersection
- 3.2.9 — Cauchy's theorem by cyclic rotation — (a)–(f)
- 3.2.18 — Coprime order and quotient index
- Additional problem — Minimal abelian normal subgroup and complement

Section 3.3 — 4 problems

- 3.3.6 — Quotient lattice of the modular group
- 3.3.7 — Two normal factors and a product quotient
- 3.3.9 — Intersecting a Sylow subgroup with a normal subgroup
- 3.3.10 — Hall subgroups under intersection and quotient

Section 3.4 — 3 problems

- 3.4.2 — All composition series of Q_8 and D_8
- 3.4.7 — A composition series through a given normal subgroup
- 3.4.11 — An abelian normal subgroup inside a solvable group

Section 3.5 — 3 problems

- 3.5.4 — Two generators for S_n
- 3.5.6 — A proper dihedral subgroup of S_4
- 3.5.10 — A composition series for A_4

Section 4.1 — 2 problems

- 4.1.7 — Blocks, primitivity, and maximal point stabilizers
- 4.1.9 — Orbits of a normal subgroup

Section 4.2 — 3 problems

- 4.2.9 — Index- p subgroups in p -groups
- 4.2.11 — Cycle structure in the regular representation
- 4.2.12 — An odd regular permutation produces an index-2 subgroup

Section 4.3 — 4 problems

- 4.3.13 — Finite groups with two conjugacy classes
- 4.3.19 — Splitting a conjugacy class in a normal subgroup
- 4.3.23 — Normalizers and conjugates of a maximal subgroup
- 4.3.24 — A finite group is not covered by conjugates of one proper subgroup

Section 4.4 — 4 problems

- 4.4.3 — Bounding automorphisms of D_8
- 4.4.8 — Characteristic subgroups — (b)–(c)
- 4.4.18 — Automorphisms of S_n are inner for $n \neq 6$
- 4.4.20 — The Thompson subgroup — (a)–(b)

Section 4.6 — 1 problems

- 4.6.4 — Three-cycles generate A_n

Section 5.1 — 1 problems

- 5.1.7 — Permuting direct-product factors

Section 5.4 — 2 problems

- 5.4.15 — Two abelian quotients give an abelian intersection quotient
- 5.4.19 — Perfect subgroups and the maximal perfect subgroup

Normal subgroups, quotient groups, and coset order

Dummit–Foote §3.1; prerequisite terminology for Week 4 isomorphism theorems.

For $N \leq G$, write $N \triangleleft G$ if $gNg^{-1} = N$ for every $g \in G$. Equivalently $gN = Ng$ for every g . The quotient group G/N consists of cosets gN with product $(gN)(hN) = ghN$, identity N , and inverse $g^{-1}N$.

Normality is exactly what makes this product independent of representatives.

For $H \leq G$, the index $[G : H]$ is the number of left cosets; H need not be normal for its index to exist. If G is finite, $[G : H] = |G|/|H|$.

The order of gN is the least positive integer k such that $g^k \in N$, and is infinite if no such k exists, because $(gN)^k = g^kN$.

Example: in $C_8 = \langle x \rangle$, the element $x\langle x^4 \rangle$ has order 4 although x has order 8. Nonexample: a reflection subgroup of D_8 is not normal, so its left cosets form a set for an action but do not form a quotient group under the displayed rule.

Subgroup lattices and Hasse diagrams

Dummit–Foote §2.5; printed pp. 66–72.

The subgroup lattice of G is its subgroups ordered by inclusion. The meet of H, K is $H \cap K$; their join is $\langle H, K \rangle$, the smallest subgroup containing both. The join need not equal the set $HK = \{hk : h \in H, k \in K\}$.

A Hasse diagram places larger subgroups higher. Connect K to H exactly when $K < H$ and no subgroup lies strictly between them; this is a cover relation. Transitive inclusions are read along paths and are not drawn as additional edges.

A partial lattice diagram may display only selected subgroups; a line then need not be a cover. An isomorphism of subgroup lattices preserves inclusion, meets, and joins, but it need not determine the group up to isomorphism: the assigned modular-group example illustrates this distinction.

Centralizers, normalizers, and the center

Dummit–Foote §2.2 and §4.3; Week 5 p. 11 and Week 6 pp. 1, 5 (PDF pages).

For a subset $S \subseteq G$,

$$C_G(S) = \{g \in G : gs = sg \text{ for all } s \in S\}, \quad N_G(S) = \{g \in G : gSg^{-1} = S\}.$$

For a single element x , write $C_G(x) = C_G(\{x\})$. The center is $Z(G) = C_G(G)$. These are subgroups. Centralizing is pointwise commutation, whereas normalizing requires preservation of the set and may permute its elements.

For $H \leq G$, $H \leq N_G(H)$, and $H \triangleleft G$ precisely when $N_G(H) = G$. A subgroup is self-normalizing if $N_G(H) = H$.

Example: in D_8 , $\langle r \rangle$ is normal, but s does not centralize r . Thus normalization does not imply centralization.

Commutators, derived subgroups, and perfectness

Dummit–Foote §3.1 Exercise 41 and §5.4; Week 5 p. 4 (PDF page).

Use the textbook convention

$$[x, y] = x^{-1}y^{-1}xy.$$

For subsets $A, B \subseteq G$, $[A, B]$ is the subgroup generated by all $[a, b]$ with $a \in A, b \in B$. In particular $G' = [G, G]$ is the commutator or derived subgroup; define $G^{(0)} = G$ and $G^{(j+1)} = (G^{(j)})'$.

A group is abelian if and only if $G' = 1$. A group is perfect if and only if $G' = G$. The trivial group is both abelian and perfect. A nontrivial abelian group is not perfect.

The lecture uses generators $xyx^{-1}y^{-1}$ for its commutator subgroup. The generated subgroup is the same: the lecture expression equals the textbook $[x^{-1}, y^{-1}]$, and inverses also range over all group elements. Preserve the textbook convention for individual commutator calculations; equality of generated subgroups does not mean the two individual expressions are equal.

Simple groups and composition series

Dummit–Foote §3.4; Week 4 p. 9 and Week 5 pp. 1–3 (PDF pages).

A group G is simple if it is nontrivial and its only normal subgroups are 1 and G . A composition series is a finite strictly increasing chain

$$1 = G_0 \triangleleft G_1 \triangleleft \dots \triangleleft G_t = G$$

whose factors G_{i+1}/G_i are nontrivial simple groups. Normality is required between consecutive terms; G_i need not be normal in G .

The factors are called composition factors. Equivalent lists have the same factor isomorphism types with multiplicity, allowing a different order. Jordan–Hölder does not say the intermediate subgroups are unique.

A cyclic group of prime order is simple. C_4 is not, because its order-2 subgroup is normal. A composition series for C_4 is $1 < C_2 < C_4$, with two C_2 factors. The trivial group has the empty composition series and no factors.

Solvable groups

Dummit–Foote §3.4; Week 5 pp. 4–5 (PDF pages).

A group G is solvable if there exists a finite chain

$$1 = G_0 \triangleleft G_1 \triangleleft \dots \triangleleft G_t = G$$

with every quotient G_{i+1}/G_i abelian. These factors need not be simple. Equivalently, the derived sequence reaches 1 after finitely many steps; this equivalence is proved in the theorem collection.

Every abelian group is solvable, using $1 \triangleleft G$, and A_4 is solvable using $1 < C_2 < V_4 < A_4$. A nonabelian perfect group is not solvable, since its derived sequence never decreases. In particular nonabelian simple groups are not solvable.

The definition is not restricted to finite groups. Finiteness is required for the additional composition-factor characterization discussed in the book.

Actions, orbits, stabilizers, and kernels

Dummit–Foote §4.1; Week 4 pp. 2–4 (PDF pages).

A left action of G on a set A is a map $(g, a) \mapsto g \cdot a$ such that $1 \cdot a = a$ and $(gh) \cdot a = g \cdot (h \cdot a)$. For $a \in A$,

$$G \cdot a = \{g \cdot a : g \in G\}, \quad G_a = \{g : g \cdot a = a\}.$$

The action is transitive if A is nonempty and it is a single orbit. Its permutation representation is $\rho : G \rightarrow S_A$, $\rho(g)(a) = g \cdot a$; the kernel is $\bigcap_{a \in A} G_a$. It is faithful if this kernel is 1.

For a subset $B \subseteq A$, the pointwise stabilizer consists of elements fixing every point of B . The setwise stabilizer is $\{g : gB = B\}$ and may permute points of B . In the lecture notation these are $\text{Stab}_G(B)$ and $\text{WStab}_G(B)$. In the assigned block problem, G_B means the setwise stabilizer. For a single point the distinction disappears.

Do not confuse an action being transitive with being faithful. A nontrivial group acting on a singleton is transitive and has kernel equal to the whole group.

Blocks and primitive permutation actions

Dummit–Foote §4.1 Exercise 7; terminology needed by the assigned homework.

For a transitive action on a nonempty set A , a block is a nonempty $B \subseteq A$ such that for every $g \in G$, $gB = B$ or $gB \cap B = \emptyset$. The action is primitive if its only blocks are singletons and A .

For $|A| > 1$, a transitive action is primitive exactly when every point stabilizer is maximal. The proof is in the assigned Exercise 4.1.7. The condition $|A| > 1$ is necessary because on a singleton the point stabilizer is the whole group and is not a maximal proper subgroup.

In the natural action of S_4 , no subset of size 2 or 3 is a block. In the square action of D_8 , opposite pairs are blocks. “Block” concerns every translate, not just invariance under one generator.

Automorphisms, inner automorphisms, and characteristic subgroups

Dummit–Foote §4.4; Week 6 pp. 8–10 (PDF pages).

An automorphism of G is an isomorphism $G \rightarrow G$. The group $\text{Aut}(G)$ uses composition as multiplication. For $g \in G$, $c_g(x) = gxg^{-1}$ is an inner automorphism; the subgroup of all these is $\text{Inn}(G)$.

A subgroup $H \leq G$ is characteristic if $\alpha(H) = H$ for every $\alpha \in \text{Aut}(G)$. A normal subgroup only requires this for inner automorphisms. Therefore characteristic implies normal, but not conversely: any order-2 subgroup of V_4 is normal, while automorphisms of V_4 permute its three such subgroups.

Characteristicity is transitive. Also H characteristic in N and N normal in G imply H normal in G , because conjugation by any $g \in G$ restricts to an automorphism of N . Replacing the first characteristicity assumption by normality can fail.

p -subgroups, Sylow subgroups, and Hall subgroups

Dummit–Foote §3.3 Exercises 9–10 and §4.5 definition; background for the assigned homework, not an additional exercise assignment.

For a prime p , a finite p -group has order a power of p . When $|G| = p^a m$ and $p \nmid m$, a Sylow p -subgroup has order p^a . In the assigned Exercise 3.3.9, such a subgroup is given; its existence need not be proved there.

A Hall subgroup H of a finite group G satisfies $\gcd(|H|, [G : H]) = 1$. Every Sylow subgroup is Hall, but a Hall subgroup may involve several primes. The trivial subgroup and the whole group are Hall. The subgroup of order 2 in C_4 is not Hall because its index is also 2.

The results needed for the assigned intersection and image problems follow from Lagrange, the product-order formula, and the isomorphism theorems. The full Sylow theorem is not required for those proofs.

Direct products, complements, and extremal subgroups

Dummit–Foote §5.1 and §5.4; Week 6 pp. 11–14 (PDF pages); additional user-supplied problem.

The external direct product $H \times K$ consists of pairs with $(h, k)(h', k') = (hh', kk')$. Its two coordinate subgroups commute elementwise. For internal subgroups $H, K \leq G$, one has $G \cong H \times K$ through $(h, k) \mapsto hk$ if $G = HK$, $H \cap K = 1$, and both are normal in G .

For $N \triangleleft G$, a complement is a subgroup H with $G = NH$ and $N \cap H = 1$. A complement need not be normal, so these conditions alone do not give a direct product.

A maximal subgroup H is proper and has no strict intermediate subgroup $H < K < G$. A minimal nontrivial normal subgroup N is nontrivial, normal, and has no $1 < L < N$ normal in G . This does not mean N has no proper nontrivial subgroups; it only restricts those normal in the ambient group.

Example: $\langle r \rangle$ and $\langle s \rangle$ in D_8 have trivial intersection and product D_8 , but $srs = r^{-1} \neq r$, so this is not $C_4 \times C_2$.

Sign and alternating groups

Dummit–Foote §3.5; Week 5 pp. 6–9 (PDF pages).

Every permutation in S_n can be written as a product of transpositions. Its sign is $+1$ when the number of transpositions is even and -1 when odd; the parity is independent of the expression. The sign map is a homomorphism $\text{sgn} : S_n \rightarrow \{1, -1\}$. For $n \geq 2$ it is surjective, and

$$A_n = \ker(\text{sgn}), \quad |A_n| = n!/2.$$

A k -cycle has sign $(-1)^{k-1}$; signs multiply over disjoint cycles. Thus a permutation is odd exactly when its disjoint-cycle decomposition has an odd number of even-length cycles.

The lecture justifies well-definedness using $D = \prod_{i < j} (x_i - x_j)$ over a field of characteristic not 2: permuting variables takes D to D or $-D$, and transpositions take it to $-D$. Applying two permutations gives multiplicativity. This polynomial argument is an explanation of sign, not an extra hypothesis on the abstract group S_n .

Example: $(1\ 2)(3\ 4)$ is even although it has order 2. For $n = 1$, $A_1 = S_1 = 1$, so the formula $n!/2$ is not applicable.

Conjugacy classes and cycle type

Dummit–Foote §4.3; Week 5 pp. 11–13 and Week 6 p. 1 (PDF pages).

Elements $x, y \in G$ are conjugate if $y = gxg^{-1}$ for some $g \in G$. The conjugacy class of x is $\{gxg^{-1} : g \in G\}$. Conjugacy is an equivalence relation: the identity gives reflexivity, inversion of the conjugating element gives symmetry, and multiplication of conjugating elements gives transitivity.

The cycle type of a permutation is the multiset of lengths of its disjoint cycles, including fixed points as 1-cycles. Two permutations in S_n are conjugate exactly when their cycle types agree. One direction follows because conjugation relabels cycles and preserves lengths. For the other, match cycles of the same lengths and send consecutive points in each first cycle to consecutive points in its matched second cycle; this defines the conjugating permutation.

Conjugacy in A_n is finer: some even S_n -classes split into two A_n -classes. The normal-subgroup class-splitting formula in 4.3.19 determines when. Normal subgroups are unions of conjugacy classes, but an arbitrary union of classes need not be a subgroup.

Cosets, Lagrange, and multiplicativity of index

Dummit–Foote §3.1 Proposition 4 and §3.2 Theorem 8; prior lecture prerequisite.

For $H \leq G$, the left cosets partition G , and each has the same cardinality as H . Thus for finite G , $|G| = [G : H]|H|$. If $H \leq K \leq G$ have finite indices, then $[G : H] = [G : K][K : H]$. In particular, the order of an element of a finite group divides the group order.

Proof

Coset equality criterion. We have $aH = bH$ exactly when $b^{-1}a \in H$. If $aH = bH$, then $a = bh$ for some $h \in H$, so $b^{-1}a = h \in H$. Conversely if $a = bh$ with $h \in H$, then $aH = bhH = bH$. If two left cosets intersect, write $ah_1 = bh_2$; then $b^{-1}a = h_2h_1^{-1} \in H$, so the cosets are equal. Every element g belongs to gH , so distinct cosets give a partition.

Sizes. Multiplication by a is a bijection $H \rightarrow aH$ with inverse multiplication by a^{-1} . For finite G , counting the partition gives $|G| = [G : H]|H|$. Apply this with $H = \langle x \rangle$ to obtain element-order divisibility.

Indices in a chain. Choose representatives g_i of left K -cosets in G , and representatives k_j of left H -cosets in K . Every element of G is some g_ik_jh , so the cosets g_ik_jH cover G . If $g_ik_jH = g_{i'}k_{j'}H$, then the two cosets lie in the same left K -coset, so $i = i'$. Cancellation then gives $k_jH = k_{j'}H$ and $j = j'$. Thus these are exactly $[G : K][K : H]$ distinct left H -cosets, proving the formula.

Products of subgroups and their orders

Dummit–Foote §3.2 Propositions 13–14 and Corollary 15; Week 4 pp. 1–3 (PDF pages).

For $H, K \leq G$, the set HK is a subgroup exactly when $HK = KH$. It is a subgroup if $H \leq N_G(K)$, in particular if $K \triangleleft G$. For finite H, K ,

$$|HK| = \frac{|H||K|}{|H \cap K|},$$

even if HK is not a subgroup.

Proof

Subgroup implies equality. Suppose HK is a subgroup. Since it contains H, K , it contains KH . For $a \in HK$, its inverse belongs to HK , say $a^{-1} = hk$. Thus $a = k^{-1}h^{-1} \in KH$, proving the other inclusion.

Equality implies subgroup. If $HK = KH$, take $a = h_1k_1, b = h_2k_2$. Then $ab^{-1} = h_1(k_1k_2^{-1})h_2^{-1}$. Rewrite the middle product $(k_1k_2^{-1})h_2^{-1}$ as h_3k_3 using $KH = HK$. Hence $ab^{-1} = h_1h_3k_3 \in HK$. The identity belongs to HK , so the subgroup test applies.

Normalization implies equality. If $hKh^{-1} = K$ for every $h \in H$, then $hk = (hkh^{-1})h \in KH$, giving $HK \subseteq KH$. Also $kh = h(h^{-1}kh) \in HK$, giving the reverse inclusion.

Count the products. Consider the surjection $H \times K \rightarrow HK, (h, k) \mapsto hk$. For a fixed expression h_0k_0 , all its preimages are precisely

$$(h_0t, t^{-1}k_0) \quad (t \in H \cap K).$$

Indeed these pairs have the same product. Conversely if $hk = h_0k_0$, then $t = h_0^{-1}h = k_0k^{-1}$ belongs to $H \cap K$, and $h = h_0t, k = t^{-1}k_0$. Thus every fiber has $|H \cap K|$ elements. Counting gives $|H||K| = |HK||H \cap K|$, as claimed. This derivation requires no subgroup assumption on HK .

Distinction: $HK = KH$ is equality of sets; it does not say each h commutes with each k .

Orbit–stabilizer and the kernel of an action

Dummit–Foote §4.1 Proposition 2; Week 4 p. 3 (PDF page).

For a group action and $a \in A$, the map $gG_a \mapsto g \cdot a$ is a bijection from the left cosets of G_a onto $G \cdot a$. Hence $|G \cdot a| = [G : G_a]$. The kernel of the permutation representation is $\bigcap_{a \in A} G_a$.

Proof

First G_a is a subgroup: the identity fixes a , and if g, h fix a , then h^{-1} fixes a and $(gh^{-1}) \cdot a = g \cdot a = a$.

For $g, h \in G$,

$$g \cdot a = h \cdot a \iff (h^{-1}g) \cdot a = a \iff h^{-1}g \in G_a \iff gG_a = hG_a.$$

This equivalence proves both well-definedness and injectivity of the displayed coset map. Surjectivity holds because every point in the orbit is by definition some $g \cdot a$. Thus it is a bijection, proving the orbit-size formula.

Each g acts by a bijection of A , with inverse the action of g^{-1} . The action law shows $\rho(gh) = \rho(g)\rho(h)$, so $\rho : G \rightarrow S_A$ is a homomorphism. Its kernel consists of the elements acting as the identity permutation, exactly those fixing every $a \in A$. This is $\bigcap_a G_a$.

First Isomorphism Theorem

Dummit–Foote §3.3 Theorem 16; Week 4 p. 5 (PDF page).

For a homomorphism $f : G \rightarrow H$, its kernel K is normal, its image is a subgroup, and

$$G/K \cong f(G), \quad gK \mapsto f(g).$$

Proof

Kernel and image. If $x \in K$ and $g \in G$, then $f(gxg^{-1}) = f(g)1f(g)^{-1} = 1$, so conjugation preserves K and $K \triangleleft G$. The subgroup tests for kernel and image follow from $f(ab^{-1}) = f(a)f(b)^{-1}$.

Well-definedness and injectivity together. For $g, h \in G$,

$$gK = hK \iff h^{-1}g \in K \iff f(h)^{-1}f(g) = 1 \iff f(g) = f(h).$$

Thus the proposed map does not depend on the coset representative, and two cosets have the same image only if they are equal.

Homomorphism and surjectivity. The coset product gives $\bar{f}((gK)(hK)) = f(gh) = f(g)f(h) = \bar{f}(gK)\bar{f}(hK)$. Every element of $f(G)$ is some $f(g)$, so the map is onto. Together with injectivity this proves the isomorphism.

Working rule: to understand a quotient, first construct a map from the original group whose kernel is the proposed denominator.

Second (diamond) Isomorphism Theorem

Dummit–Foote §3.3 Theorem 18; Week 4 p. 6 (PDF page).

If $H, K \leq G$ and $H \leq N_G(K)$, then $HK \leq G$, $K \triangleleft HK$, $H \cap K \triangleleft H$, and

$$H/(H \cap K) \cong HK/K.$$

In particular these conclusions hold when $K \triangleleft G$.

Proof

The product theorem gives $HK \leq G$. An element $hk \in HK$ normalizes K , because h does by hypothesis and k does because it is in K . Thus $K \triangleleft HK$.

Define $f : H \rightarrow HK/K$ by $f(h) = hK$. This is a homomorphism as a restriction of the quotient projection. Every coset in HK/K is $(hk)K = hK$, so f is surjective. Its kernel is the set of $h \in H$ lying in K , namely $H \cap K$. Kernels are normal, so $H \cap K \triangleleft H$. The First Isomorphism Theorem gives the desired isomorphism.

The diamond has top HK , middle vertices H, K , and bottom $H \cap K$. It summarizes the two sides identified by the map, but does not assert the displayed inclusions are covers.

Third Isomorphism Theorem

Dummit–Foote §3.3 Theorem 19; Week 4 p. 7 (PDF page).

If $H, K \triangleleft G$ with $H \leq K$, then $K/H \triangleleft G/H$ and

$$(G/H)/(K/H) \cong G/K.$$

The natural isomorphism sends $(gH)(K/H)$ to gK .

Proof

Define $f : G/H \rightarrow G/K$ by $f(gH) = gK$. If $gH = g'H$, then $g'^{-1}g \in H \leq K$, so $gK = g'K$. This verifies representative independence.

The map is a homomorphism because $f(ghH) = ghK = (gK)(hK)$. It is surjective because every gK is the image of gH . Its kernel consists of cosets gH with $g \in K$, exactly K/H . Being a kernel, K/H is normal in G/H . The First Isomorphism Theorem applied to f produces the stated quotient-of-a-quotient isomorphism.

Working rule: define the obvious map and check its kernel; avoid treating quotient notation as literal fraction cancellation.

Subgroup correspondence (Lattice Isomorphism Theorem)

Dummit–Foote §3.3 Theorem 20; Week 4 p. 8 (PDF page).

Let $N \triangleleft G$ and $\pi : G \rightarrow G/N$ be projection. Subgroups H of G containing N correspond bijectively to subgroups $\bar{H}$ of G/N by

$$H \mapsto H/N, \quad \bar{H} \mapsto \pi^{-1}(\bar{H}).$$

This correspondence preserves inclusion, intersections, generated joins, and indices. For $N \leq H \leq K$, $H \triangleleft K$ exactly when $H/N \triangleleft K/N$, and then $(K/N)/(H/N) \cong K/H$.

Proof

The constructions give subgroups. Images and preimages of subgroups under homomorphisms are subgroups: the subgroup test follows by applying the map to ab^{-1} . The preimage always contains N , because N maps to the identity.

The constructions are inverse. For $N \leq H$, an element g belongs to $\pi^{-1}(\pi(H))$ exactly when $gN = hN$ for some $h \in H$. Then $h^{-1}g \in N \leq H$, so $g \in H$. The reverse inclusion is immediate. Conversely, since π is onto, every point of a subgroup $\bar{H} \leq G/N$ has a preimage, so $\pi(\pi^{-1}(\bar{H})) = \bar{H}$.

Inclusion and intersections. Inclusion is preserved by images and preimages. If H, K contain N , then $gN \in (H/N) \cap (K/N)$ implies $g \in H \cap K$ by the inverse property, proving $(H \cap K)/N = (H/N) \cap (K/N)$. The reverse inclusion is also direct.

Joins. The image under a homomorphism of a word in elements of H, K is the corresponding word in their images. Conversely any word in the images lifts to that word in H, K . Hence $\pi(\langle H, K \rangle) = \langle \pi(H), \pi(K) \rangle$.

Indices. For $N \leq H \leq K$, the map from left H -cosets in K to left H/N -cosets in K/N , given by $kH \mapsto (kN)(H/N)$, is a bijection: equality of the target cosets says $\pi(k'^{-1}k) \in H/N$, equivalently $k'^{-1}k \in H$. Surjectivity follows from surjectivity of $K \rightarrow K/N$. Therefore $[K : H] = [K/N : H/N]$.

Normality. If $H \triangleleft K$, conjugation by kN sends H/N to itself. Conversely normality of H/N in K/N implies $\pi(khk^{-1}) \in H/N$ for $k \in K, h \in H$, so $khk^{-1} \in H$. Applying the statement to k^{-1}

gives equality. Finally, the map $K/N \rightarrow K/H, kN \mapsto kH$, is well defined, onto, and has kernel H/N , so the First Isomorphism Theorem gives the last assertion.

THEOREM • cu:algebra:midterm-2:theorems:th:commutator-criterion

Commutator criterion for abelian quotients

Dummit–Foote §3.1 Exercises 40–41 and §5.4; Week 5 p. 4 and Week 6 p. 10 (PDF pages).

For $N \triangleleft G$, G/N is abelian if and only if $G' \leq N$. The subgroup G' is characteristic in G . If $N \triangleleft G$, then $N' \triangleleft G$.

Proof

For any $x, y \in G$, the cosets xN, yN commute exactly when $(xN)^{-1}(yN)^{-1}(xN)(yN) = N$, equivalently $[x, y] \in N$. Thus all quotient cosets commute exactly when all generating commutators of G' lie in N , equivalently $G' \leq N$.

For $\alpha \in \text{Aut}(G)$,

$$\alpha([x, y]) = [\alpha(x), \alpha(y)].$$

Since α is bijective, it permutes the set of all commutators. It therefore preserves the subgroup they generate, proving G' characteristic.

If $N \triangleleft G$, conjugation by any $g \in G$ restricts to an automorphism of N . Applying the preceding argument inside N shows that conjugation preserves N' . Hence $N' \triangleleft G$. This last conclusion can also be checked directly on commutator generators, as in the solution of 3.4.11.

Derived sequence criterion and solvable extensions

Dummit–Foote §3.4 and §5.4; Week 5 pp. 4–5 (PDF pages).

A group G is solvable if and only if $G^{(t)} = 1$ for some finite t . Subgroups and quotient groups of solvable groups are solvable. If $N \triangleleft G$, then G is solvable if and only if both N and G/N are solvable.

Proof

Solvable series implies derived termination. Write a solvable series downward as $G = G_0 \supseteq G_1 \supseteq \dots \supseteq G_t = 1$, each adjacent quotient abelian. The commutator criterion gives $G'_i \leq G_{i+1}$. Inductively $G^{(i)} \leq G_i$: the base case is equality, and commutators of elements of $G^{(i)} \leq G_i$ lie in $G'_i \leq G_{i+1}$. Thus $G^{(t)} = 1$.

Derived termination implies a solvable series. Each $G^{(i+1)}$ is normal in $G^{(i)}$, and the quotient by its commutator subgroup is abelian. Hence the finite derived chain ending at 1 is a solvable series, read upward. Repeated terms can be omitted.

Subgroups. If $H \leq G$, then $H^{(i)} \leq G^{(i)}$ by induction: it holds initially, and generating commutators preserves inclusion. Termination for G therefore implies termination for H .

Quotients. A surjective homomorphism $f : G \rightarrow Q$ satisfies $f(G') = Q'$, because it sends every commutator to a commutator, and every commutator in Q has preimages in G . Induction gives $f(G^{(i)}) = Q^{(i)}$. Thus quotients of solvable groups are solvable.

Extensions. If N and G/N are solvable, choose s, t with $(G/N)^{(s)} = 1$ and $N^{(t)} = 1$. Projection then gives $G^{(s)} \leq N$. Applying the subgroup-inclusion observation repeatedly yields

$$G^{(s+t)} = (G^{(s)})^{(t)} \leq N^{(t)} = 1.$$

So G is solvable. Conversely, if G is solvable, the subgroup and quotient results already proved apply to N and G/N . No finite-order assumption is required.

Conjugacy classes, the class equation, and p -group centers

Dummit–Foote §4.3 Theorem 7 and Theorem 8/Corollary 9; Week 6 p. 5 (PDF page).

For finite G , the conjugacy class of x has size $[G : C_G(x)]$. If $x_1, \dots, x_r$ represent its noncentral classes, then

$$|G| = |Z(G)| + \sum_{i=1}^r [G : C_G(x_i)].$$

Every nontrivial finite p -group has nontrivial center. Every group of order p^2 is abelian.

Proof

Conjugation defines an action $g \cdot x = gxg^{-1}$. Its stabilizer at x consists exactly of elements commuting with x , namely $C_G(x)$. Orbit–stabilizer gives the class size. An element has singleton class precisely when it commutes with every element, so precisely when it lies in $Z(G)$. Partitioning G into singleton central classes and the remaining classes gives the displayed equation.

For $|G| = p^a$, every noncentral centralizer is proper, so each corresponding index is a power of p greater than 1 and hence divisible by p . Reducing the class equation modulo p gives $|Z(G)| \equiv 0 \pmod{p}$. The center contains the identity and thus is nonempty. Its positive order is a multiple of p , so at least p , proving it nontrivial.

For $|G| = p^2$, the center has order p or p^2 by Lagrange. In the latter case G is abelian. In the former case $G/Z(G)$ has order p and is cyclic, so the central-quotient argument from Exercise 3.1.36 again makes G abelian. Consequently the center is actually all of G .

Technique: noncentral orbits have sizes divisible by p , so singleton orbits account for the residue.

Coset actions, core, Cayley, and smallest-prime index

Dummit–Foote §4.2 Theorem 3 and Corollaries 4–5; Week 5 p. 10 (PDF page).

For $H \leq G$, left multiplication on the left cosets of H gives $\rho : G \rightarrow S_{G/H}$ with kernel

$$\text{core}_G(H) = \bigcap_{g \in G} gHg^{-1} \leq H.$$

For $H = 1$ this is faithful (Cayley's theorem). If G is finite and p is the smallest prime dividing $|G|$, every subgroup of index p is normal.

Proof

The coset action. If $aH = bH$, multiplying both cosets by g gives $gaH = gbH$, so $g \cdot (aH) = gaH$ is well defined. Identity and associativity give the action laws. An element k fixes every left coset exactly when $kgH = gH$ for every g , equivalently $g^{-1}kg \in H$ for every g , or $k \in \bigcap_g gHg^{-1}$. This is the kernel, and taking $g = 1$ shows it lies in H .

Cayley. For $H = 1$, the kernel is contained in 1 and is therefore trivial. The First Isomorphism Theorem identifies G with the image, a subgroup of the symmetric group on its underlying set.

Smallest-prime index. If $[G : H] = p$, the coset image lies in S_p . Write $K = \ker \rho \leq H$ and $k = [H : K]$. The image order is $[G : K] = pk$ by index multiplicativity, and divides $p!$, so k divides $(p - 1)!$. Also k divides $|G|$, since it divides $|H|$. Any prime divisor of k would thus be at least p by minimality of p , while any prime divisor of $(p - 1)!$ is smaller than p . This is impossible. Therefore $k = 1$, so $H = K$ and H is normal.

The prime being the smallest matters. A maximal subgroup with composite index need not be normal, and an arbitrary prime-index subgroup need not be normal either.

Inner automorphisms and characteristic-to-normal transfer

Dummit–Foote §4.4; Week 6 pp. 8–10 (PDF pages).

One has $\mathbf{Inn}(G) \cong G/Z(G)$ and $\mathbf{Inn}(G) \triangleleft \mathbf{Aut}(G)$. Characteristic subgroups are normal; characteristicity is transitive; and a characteristic subgroup of a normal subgroup is normal in the whole group.

Proof

The map $c : G \rightarrow \mathbf{Aut}(G)$, $c(g) = c_g$, is a homomorphism because $c_g c_h(x) = ghx(gh)^{-1} = c_{gh}(x)$. Its image is by definition $\mathbf{Inn}(G)$. Its kernel consists of those g commuting with every x , namely $Z(G)$. The First Isomorphism Theorem gives $G/Z(G) \cong \mathbf{Inn}(G)$.

For $\alpha \in \mathbf{Aut}(G)$, direct evaluation at x gives

$$\alpha c_g \alpha^{-1}(x) = \alpha(g)x\alpha(g)^{-1} = c_{\alpha(g)}(x).$$

Thus automorphisms conjugate inner automorphisms to inner automorphisms, proving $\mathbf{Inn}(G)$ normal in $\mathbf{Aut}(G)$.

If H is characteristic in G , all inner automorphisms preserve it, so H is normal. If H is characteristic in K and K characteristic in G , every automorphism of G restricts to an automorphism of K and hence preserves H , proving transitivity. If instead $K \triangleleft G$, every conjugation by $g \in G$ restricts to an automorphism of K ; characteristicity of H in K makes this restriction preserve H , proving $H \triangleleft G$.

Recognizing an internal direct product

Dummit–Foote §5.4 Theorem 9; Week 6 p. 12 (PDF page).

If $H, K \triangleleft G$, $H \cap K = 1$, and $G = HK$, then

$$H \times K \cong G, \quad (h, k) \mapsto hk.$$

In particular, normal subgroups with trivial intersection commute elementwise.

Proof

Step 1: show the two factors commute. For $h \in H, k \in K$, the commutator $h^{-1}k^{-1}hk$ lies in H , since $k^{-1}hk \in H$ by normality. It also lies in K , since $h^{-1}k^{-1}h \in K$. Thus it is in $H \cap K = 1$, giving $hk = kh$.

Step 2: define the map and check multiplication. Put $f(h, k) = hk$. Using the commutation just proved,

$$f((h, k)(h', k')) = hh'kk' = kh'h'k' = f(h, k)f(h', k').$$

So f is a homomorphism. The assumption $G = HK$ is exactly its surjectivity.

Step 3: prove injectivity. If $f(h, k) = 1$, then $h = k^{-1}$ lies in $H \cap K$, so both h and k are identity. The kernel is trivial, making f injective. Thus it is an isomorphism.

Both normality hypotheses are used in Step 1. If just one factor is normal, a complement may exist without giving a direct product.

Existence and uniqueness of composition factors

Dummit–Foote §3.4 Theorem 22; Week 5 pp. 1–3 (PDF pages).

Every finite group has a composition series. Any two composition series have the same length and the same multiset of simple factor isomorphism types (Jordan–Hölder). The order of those factors and the intermediate subgroups may differ.

Proof

Existence by decreasing orders. If $G = 1$, use the empty series. Otherwise choose a maximal proper normal subgroup $M \triangleleft G$. It exists because there are finitely many subgroups and at least the proper normal subgroup 1 . The quotient G/M is nontrivial and simple: a nontrivial proper normal subgroup of it would pull back to a proper normal subgroup strictly between M and G , contradicting maximality. Continue the same construction inside M . Each chosen subgroup has smaller finite order, so the process eventually reaches 1 . Read the chain upward to obtain a composition series.

Uniqueness by induction on $|G|$. The trivial group has only the empty series. Assume uniqueness for all groups of order smaller than $|G|$, and consider two composition series of G , with penultimate terms A, B . Each is a maximal proper normal subgroup of G : the corresponding top factor is simple, and subgroup correspondence forbids any proper intermediate normal subgroup.

If $A = B$, the portions of both series inside this same subgroup have the same factor multiset by induction. Adding the common final factor G/A proves the result.

Now suppose $A \neq B$. Neither can contain the other, since both are maximal proper normal subgroups. The subgroup AB is normal in G , because conjugation preserves both factors, and it strictly contains A . Hence $AB = G$. Let $D = A \cap B$, which is normal in G . The second isomorphism theorem gives

$$A/D \cong G/B, \quad B/D \cong G/A.$$

Both quotients are nontrivial simple, so D is a maximal proper normal subgroup of each of A, B .

Choose any composition series of D , whose existence was already proved. Appending A to it gives a composition series of A , since its new top factor is A/D . Appending B instead gives one of B with top factor B/D . The induction hypothesis applies to A and B since both have smaller order than G . It says that the original first series below G has, inside A , the factors of D together with A/D , and the original second has, inside B , the factors of D together with B/D .

Thus the factors of the first whole series are the factors of D , then A/D , then G/A . Those of the second are the factors of D , then B/D , then G/B . The two displayed isomorphisms identify the last two factors in opposite order. Therefore the multisets agree, including multiplicities, and their counts agree too. This completes the induction.

What uniqueness does not mean. It does not make the series itself unique or reconstruct G from its factors. The assigned Q_8, D_8 examples have the same three C_2 factors but are different groups.

Simplicity of A_n for $n \geq 5$

Dummit–Foote §4.6 Theorem 23; Week 6 pp. 1–4 (PDF pages).

For every $n \geq 5$, A_n is nonabelian simple. Its generation by 3-cycles is proved separately in the assigned Exercise 4.6.4.

Proof

Base case: A_5 . The possible even cycle types on five points are identity, a double transposition, a 3-cycle, and a 5-cycle. In S_5 their class sizes are 1, 15, 20, 24, respectively: there are 5 choices of the fixed point and 3 pairings for double transpositions, $\binom{5}{3} \cdot 2 = 20$ 3-cycles, and $4! = 24$ 5-cycles.

The double-transposition class remains one A_5 -class because its centralizer contains an odd permutation, for example $(1\ 2)$ for $(1\ 2)(3\ 4)$. The 3-cycle class also remains one class because the transposition of its two fixed points centralizes it. To justify this criterion, the splitting formula in 4.3.19 says the number of A_5 -classes is $[S_5 : A_5 C_{S_5}(x)]$, which is 1 exactly when the centralizer contains an odd element.

A permutation commuting with the 5-cycle $c = (1\ 2\ 3\ 4\ 5)$ is determined by its value at 1: commutation forces its value at $c^j(1)$ to be c^j of that value. The five choices give exactly the powers of c , all even. Thus its centralizer is contained in A_5 ; the class splits into two equally sized A_5 -classes, each of size 12.

Hence the A_5 class sizes are 1, 15, 20, 12, 12. A normal subgroup is a union of whole conjugacy classes and contains the identity. Its possible order is therefore 1 plus a subset sum of 15, 20, 12, 12. The distinct possibilities are

$$1, 13, 16, 21, 25, 28, 33, 36, 40, 45, 48, 60.$$

By Lagrange its order must also divide 60. Among those listed, only 1 and 60 do. Thus A_5 has no nontrivial proper normal subgroup and is simple.

Induction step. Let $n \geq 6$ and assume A_{n-1} simple. Take a nontrivial $H \triangleleft A_n$. For each point i , let S_i be its stabilizer in A_n . Restricting to the other $n - 1$ points identifies $S_i \cong A_{n-1}$, so S_i is simple.

If some $1 \neq h \in H$ fixes i , then $H \cap S_i$ is a nontrivial normal subgroup of S_i : it is normalized by S_i because H is normal in A_n . Simplicity forces $S_i \leq H$. For every j , an even permutation sends i to j (for $i \neq j$ use a 3-cycle through them and a third point). Conjugating then gives $S_j \leq H$. Every 3-

cycle fixes some point when $n \geq 6$, so it belongs to some S_j . As the 3-cycles generate A_n , we obtain $H = A_n$.

It remains to rule out a proper nontrivial H whose nonidentity elements fix no point. In such a subgroup, if $h, h' \in H$ agree at any point a , then $h^{-1}h'$ fixes a , so it must be identity; hence $h = h'$.

Take $1 \neq h \in H$. If one cycle of h has length at least 3, name consecutive points a_1, a_2, a_3 in it. Choose distinct d, e outside these three and set $\alpha = (a_3 d e) \in A_n$. It fixes a_1, a_2 but moves a_3 . The conjugate $h' = \alpha h \alpha^{-1}$ lies in H , agrees with h at a_1 (both send it to a_2), and differs at a_2 (its image is $\alpha(a_3) \neq a_3$). This contradicts the agreement property.

Thus h consists entirely of disjoint transpositions and has no fixed points. Because h is even, the number of transpositions is even; with $n \geq 6$, this forces at least four transpositions and in particular at least three. Name three as $(a_1 a_2), (a_3 a_4), (a_5 a_6)$. Let $\alpha = (a_1 a_2)(a_3 a_5) \in A_n$. The conjugate h' again lies in H . It still contains $(a_1 a_2)$, so agrees with h at a_1 , but it sends a_3 to a_6 instead of a_4 , since its relabeled pairs include $(a_5 a_4)$ and $(a_3 a_6)$. Hence $h' \neq h$, another contradiction. There is no proper nontrivial H , completing simplicity.

Finally A_n is nonabelian: the 3-cycles $(1 2 3)$ and $(1 2 4)$ do not commute (their products send 1 to different points). This works for every $n \geq 5$.

Technique: the lecture proof first uses the A_5 class equation, then point stabilizers and conjugates agreeing at one point. The key induction step needs both normality of H and simplicity of each smaller point stabilizer.

Orbit counting by fixed points

Week 6 pp. 6–7 (PDF pages), labeled Cauchy–Frobenius in the notes; elementary double-counting derivation.

If a finite group G acts on a finite set A , the number of orbits is

$$\frac{1}{|G|} \sum_{g \in G} |\text{Fix}_A(g)|.$$

If the action is transitive and $|A| > 1$, some element has no fixed point.

Proof

Count the finite set $X = \{(g, a) \in G \times A : g \cdot a = a\}$. Counting first by g gives $|X| = \sum_g |\text{Fix}_A(g)|$. Counting by a gives $|X| = \sum_a |G_a|$.

Within an orbit O , orbit–stabilizer gives $|G_a| = |G|/|O|$ for every $a \in O$. Hence

$$\sum_{a \in O} |G_a| = |O| \frac{|G|}{|O|} = |G|.$$

If there are r orbits, summing yields $|X| = r|G|$. Equating the two counts proves the formula.

For a transitive action $r = 1$, so $\sum_g |\text{Fix}_A(g)| = |G|$. The identity contributes $|A| > 1$. If every other element contributed at least one fixed point, the total would be at least $|A| + (|G| - 1) > |G|$, a contradiction. Thus some element contributes zero.

Scope: this result supplies lecture context for the conjugate-union problem; the assigned solution of 4.3.24 uses the book's maximal-subgroup argument instead.

Cauchy's theorem

Dummit–Foote §3.2 Exercise 9; detailed cyclic-rotation proof using orbit–stabilizer from Week 4 p. 3.

If a prime p divides the order of a finite group G , then G contains an element of order p .

Proof

Use the set of tuples $S = \{(x_1, \dots, x_p) : x_1 \cdots x_p = 1\}$. Its first $p - 1$ entries are arbitrary and determine the last, so $|S| = |G|^{p-1}$ is divisible by p .

Cyclic rotation preserves S : $x_2 \cdots x_p x_1 = x_1^{-1}(x_1 \cdots x_p)x_1 = 1$. The cyclic group of order p therefore acts on S . Every orbit has size 1 or p , by orbit–stabilizer and primality. A singleton is exactly a constant tuple $(x, \dots, x)$ with $x^p = 1$.

Let k count those singletons. Counting the other orbits in multiples of p gives $k \equiv |S| \equiv 0 \pmod{p}$. The identity tuple shows $k \geq 1$, so $k \geq p > 1$. Another singleton therefore gives $x \neq 1$ with $x^p = 1$. Its order divides p and is not 1, hence is p .

The six-part assigned exercise supplies every detail of this argument: 3.2.9 — Cauchy's theorem by cyclic rotation — (a)–(f) — Detailed solution. There is no commutativity assumption.

Technique — Define a map before manipulating quotients

Homework-aligned proof guidance; internally derived from the supplied textbook and lecture notes.

When the target resembles a quotient, begin with a homomorphism from the original group. State its domain and codomain, prove it preserves products, solve the kernel equation, and identify its image. Then invoke the First Isomorphism Theorem.

For a map out of a quotient, verify that changing the representative does not change the image. Equivalently, when starting with $f : G \rightarrow H$, check the proposed denominator is contained in $\ker f$.

Examples: the triangular-matrix projections in 3.1.11; the map $g \mapsto (gM, gN)$ in 3.3.7; and the sign of the regular representation in 4.2.12.

Common mistake: two surjective coordinate maps do not necessarily make a map onto their product. In 3.3.7, $G = MN$ is the hypothesis proving joint surjectivity.

Dependency: First Isomorphism Theorem.

Technique — Prove invariance on generators, then on words

Homework-aligned proof guidance; internally derived from the supplied textbook and lecture notes.

If $N = \langle S \rangle$, conjugation or an automorphism respects products and inverses. Thus checking the images of S lie in N proves the image of N is contained in N . To obtain equality, also apply the inverse transformation, or use equal finite cardinalities when that hypothesis is available.

Use this for 3.1.28 and 3.1.41, the hint in 3.4.11, and characteristicity of the Thompson or commutator subgroup. For characteristicity, show every automorphism preserves the family defining the subgroup, not just inner automorphisms.

Common mistake: for an infinite subgroup, a single conjugation mapping generators into the subgroup need not normalize it. Exercise 3.1.28 has an explicit matrix counterexample.

Dependency: Commutator criterion for abelian quotients.

Technique — Two divisibility constraints on one subgroup or image

Homework-aligned proof guidance; internally derived from the supplied textbook and lecture notes.

An intersection $H \cap K$ is a subgroup of both factors, so its order divides both when they are finite. Similarly, the image of H in G/N has order dividing both $|H|$ and $[G : N]$.

For prime-power questions, introduce an unknown exponent, e.g. $|P \cap N| = p^c$. Lagrange gives one inequality, and the product-order formula plus divisibility of $|PN|$ gives the other. For Hall subgroups, show the new order and index divide the original coprime order and index.

Examples: 3.2.8, 3.2.18, 3.3.9, and 3.3.10.

Common mistake: trivial intersection need not imply coprime orders; V_4 supplies a counterexample.

Dependencies: Cosets, Lagrange, and multiplicativity of index and Products of subgroups and their orders.

Technique — Choose the set, identify the stabilizer, then count

Homework-aligned proof guidance; internally derived from the supplied textbook and lecture notes.

Write down the acted-on set before invoking orbit–stabilizer. The same group can act on points, cosets, conjugacy classes, subgroups, or orbits of a normal subgroup, with different stabilizers.

- On points: G_a .
- On an element by conjugation: $C_G(x)$.
- On a subgroup by conjugation: $N_G(H)$.
- On the H -orbit of a point, when $H \triangleleft G$: HG_a .
- On the H -conjugacy class of x , when $H \triangleleft G$: $HC_G(x)$.

The last two stabilizer formulas require proof; do not infer them from notation. They explain 4.1.9 and 4.3.19. A coset action instead bounds a quotient of G by a subgroup of S_n , as in 4.2.9.

Dependency: Orbit–stabilizer and the kernel of an action.

Technique — Conjugate supports and pair transpositions

Homework-aligned proof guidance; internally derived from the supplied textbook and lecture notes.

Permutation products act right to left. Conjugation relabels cycles:

$$\tau(a_1 \cdots a_k)\tau^{-1} = (\tau(a_1) \cdots \tau(a_k)).$$

To prove this, evaluate both sides at each $\tau(a_i)$; both send it to $\tau(a_{i+1})$, with indices cyclic, and fix the remaining points.

Generation problems often start with one transposition and conjugate it to obtain adjacent or star transpositions. For A_n , group an even-length transposition expression into pairs. Distinguish identical pairs, pairs sharing one point, and disjoint pairs.

For automorphisms, orders alone may not distinguish transpositions from double or triple transpositions. Class size and intersections of supports supply the extra constraints in 4.4.18.

Common mistake: “even permutation” concerns its sign, not whether its element order is even. A double transposition is even and has order 2.

See 4.6.4 — Three-cycles generate A_n and 4.4.18 — Automorphisms of S_n are inner for $n \neq 6$.

Technique — Build below and above a normal subgroup

Homework-aligned proof guidance; internally derived from the supplied textbook and lecture notes.

For a composition series through $H \triangleleft G$, take a series inside H and a series in G/H , pull the quotient series back, and join the two chains at H . Verify both adjacent normality and simplicity of every factor.

To find an abelian normal subgroup in a solvable group, form the derived sequence of the given normal subgroup and take its last nontrivial term. Explain separately why all terms remain normal in the ambient group and why the sequence terminates.

For Jordan–Hölder, if two penultimate terms differ, intersect them. The diamond theorem identifies the two extra simple factors in opposite order.

Common mistake: a composition-series term may be normal only in its immediate overgroup. Demanding normality in all of G would wrongly discard several valid D_8 series.

Dependencies: Subgroup correspondence (Lattice Isomorphism Theorem) and Derived sequence criterion and solvable extensions.

Technique — Minimality and maximality turn containment into a dichotomy

Homework-aligned proof guidance; internally derived from the supplied textbook and lecture notes.

For a minimal nontrivial normal subgroup N , first prove the subgroup you have constructed is normal in the entire ambient group and lies in N . Only then may you conclude it is 1 or N .

For a maximal proper subgroup M , first exhibit an intermediate subgroup containing M . Its only possible values are M and G . Examples are $N_G(M)$ in 4.3.23 and the setwise stabilizer of a block in 4.1.7.

To show a complement is maximal, take an arbitrary overgroup $K > H$, produce a nonidentity element of $N \cap K$ by writing $k = nh$, and reuse the minimal-normal-subgroup argument.

Common mistake: normality in N is weaker than normality in G . In the additional problem, abelianness of N and the factorization $G = NH$ together supply the missing ambient normality.

See Additional problem — Minimal abelian normal subgroup and complement.

Technique — Exhaust maximal subgroups before drawing a lattice

Homework-aligned proof guidance; internally derived from the supplied textbook and lecture notes.

For the order-8 and order-16 lattice exercises, first name every maximal subgroup. Build its internal lattice, then identify subgroups shared with other maximal branches. Use element orders to verify the cyclic chains and common involutions.

After constructing the diagram, give a complete cover list. Confirm the total vertex count, and justify that no subgroup is missing: the supplied assertion that every proper subgroup lies in one of the maximal subgroups is enough once their internal lattices are exhausted.

For a quotient lattice, retain only subgroups containing the kernel, replace each by its image quotient, and divide finite subgroup orders by the kernel order. Determine the quotient's group type separately by a map from a familiar group.

Common mistake: the modular group and $C_2 \times C_8$ have the same lattice; matching a lattice therefore does not establish a group isomorphism.

Dependency: Subgroup correspondence (Lattice Isomorphism Theorem).

2.5.12 — Subgroup lattice of $C_2 \times C_4$

Dummit–Foote, §2.5, Exercise 12; printed pp. 71–72 (PDF pp. 85–86). Statement normalized from the supplied book; solution independently derived.

Let $A = \langle a, b \mid a^2 = b^4 = 1, ab = ba \rangle \cong C_2 \times C_4$. Its three subgroups of order 4 are $\langle a, b^2 \rangle$, $\langle b \rangle$, and $\langle ab \rangle$, and every proper subgroup lies in one of these. Draw the complete subgroup lattice, expressing every subgroup using at most two generators.

Solution

Step 1: locate the subgroups of order 2. Every element is uniquely $a^i b^j$ with $i \in \{0, 1\}$ and $0 \leq j < 4$. Since $(a^i b^j)^2 = b^{2j}$, an element has order dividing 2 precisely when j is even. The three nonidentity involutions are a, b^2, ab^2 . Thus the order-2 subgroups are

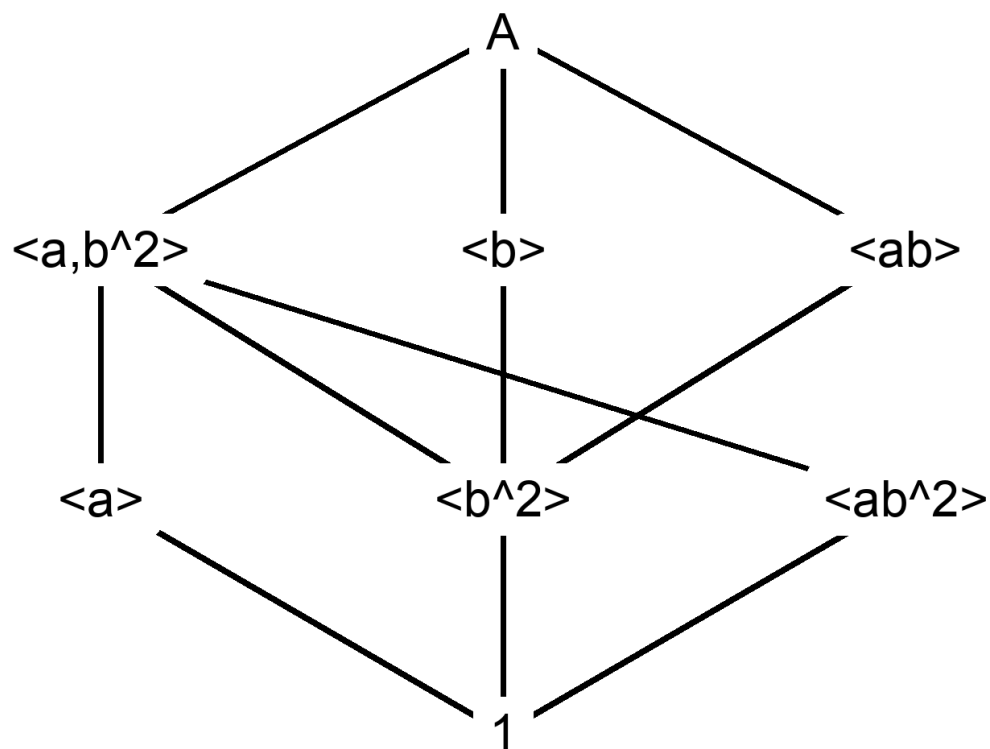
$$X = \langle a \rangle, \quad Y = \langle b^2 \rangle, \quad Z = \langle ab^2 \rangle.$$

Each has exactly one nonidentity element, so these subgroups are distinct.

Step 2: determine the containment edges. Put $V = \langle a, b^2 \rangle$, $B = \langle b \rangle$, and $D = \langle ab \rangle$. The subgroup V contains X, Y, Z . The cyclic group B has the unique subgroup Y of order 2. Since $(ab)^2 = b^2$, the cyclic group D also has unique order-2 subgroup Y .

The complete Hasse diagram is given below. Higher vertices are larger subgroups, and each line is a cover relation. There are eight vertices in total. Its edge list, useful for checking the drawing, is: A covers V, B, D ; V covers X, Y, Z ; B, D each cover Y ; and X, Y, Z each cover 1 .

Step 3: completeness. By Lagrange, a proper subgroup has order 1, 2, or 4. We found every subgroup of order 2 by listing all involutions. The three order-4 subgroups and the fact that they contain every proper subgroup are supplied in the problem; their own subgroups have now all been listed. Nothing is missing.



Technique: list element orders, then record covers rather than every transitive inclusion. A lattice specifies containments; the fact that two cyclic subgroups share an involution is essential to its shape.

Foundations: Subgroup lattices and Hasse diagrams, Cosets, Lagrange, and multiplicativity of index.

2.5.13 — Subgroup lattice of $C_2 \times C_8$

Dummit–Foote, §2.5, Exercise 13; printed p. 72 (PDF p. 86). Statement normalized from the supplied book; solution independently derived.

Let $G = \langle x, y \mid x^2 = y^8 = 1, xy = yx \rangle \cong C_2 \times C_8$. Its three order-8 subgroups are $\langle x, y^2 \rangle$, $\langle y \rangle$, and $\langle xy \rangle$, and every proper subgroup lies in one of them. Draw the complete subgroup lattice with at most two generators per subgroup.

Solution

Step 1: the three maximal subgroups. Write

$$A = \langle x, y^2 \rangle, \quad B = \langle y \rangle, \quad C = \langle xy \rangle.$$

They all have order 8. The group A is $C_2 \times C_4$, so the preceding exercise gives its complete lattice after substituting $a = x, b = y^2$.

Step 2: all order-4 and order-2 subgroups. The order-4 subgroups of A are

$$D = \langle x, y^4 \rangle, \quad E = \langle y^2 \rangle, \quad F = \langle xy^2 \rangle.$$

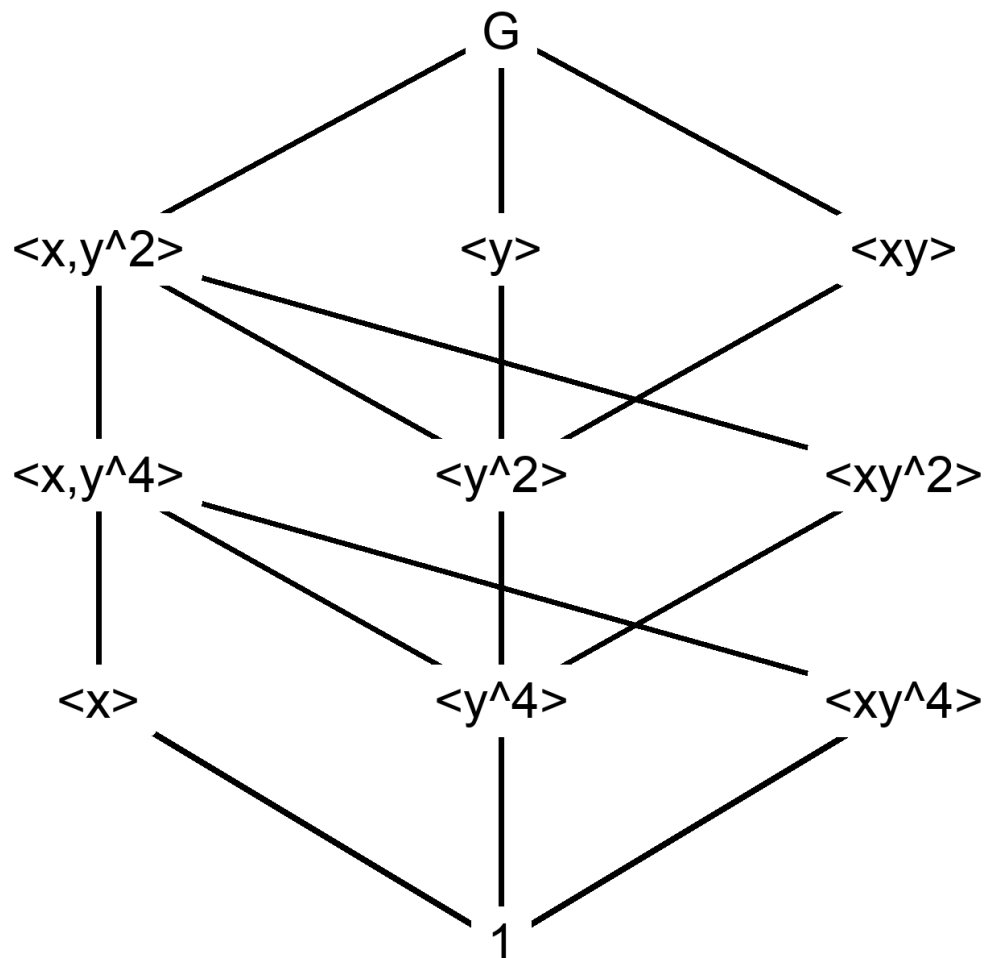
The order-2 subgroups are

$$X = \langle x \rangle, \quad Y = \langle y^4 \rangle, \quad Z = \langle xy^4 \rangle.$$

The cyclic group B has the chain $1 < Y < E < B$. Also $(xy)^2 = y^2$, so C has the chain $1 < Y < E < C$. These add no vertices below order 8 beyond those already inside A .

The complete diagram is below. The cover relations are: G covers A, B, C ; A covers D, E, F ; B, C cover E ; D covers X, Y, Z ; E, F cover Y ; X, Y, Z cover 1 . There are eleven subgroups.

Step 3: completeness. Every proper subgroup is contained in A, B , or C by the supplied hypothesis. We have exhausted the lattice of A and both cyclic chains. Taking their union therefore gives every subgroup of G .



Technique: build the lattice from maximal subgroups and identify shared subgroups before drawing separate branches.

Foundations: Subgroup lattices and Hasse diagrams, Cosets, Lagrange, and multiplicativity of index.

2.5.14 — Same lattice, different groups

Dummit–Foote, §2.5, Exercise 14; printed p. 72 (PDF p. 86). Statement normalized from the supplied book; solution independently derived.

Let $M = \langle u, v \mid u^2 = v^8 = 1, vu = uv^5 \rangle$ be the modular group of order 16. Its three order-8 subgroups are $\langle u, v^2 \rangle$, $\langle v \rangle$, and $\langle uv \rangle$, and every proper subgroup is contained in one of them. Prove that these have types $C_2 \times C_4$, C_8 , C_8 , respectively. Show that M has the same subgroup lattice as $C_2 \times C_8$ but is not isomorphic to it.

Solution

Step 1: a useful conjugation rule. The relation $vu = uv^5$ gives $uvu = v^5$ because $u^2 = 1$. Thus

$$uv^j u = v^{5j}$$

for every integer j . In particular $uv^2 u = v^{10} = v^2$, so u commutes with v^2 .

Step 2: identify the maximal subgroups. The commuting generators u and v^2 have orders 2 and 4. Their cyclic subgroups intersect trivially: if u belonged to $\langle v \rangle$, then u, v would generate a group of at most order 8, contrary to $|M| = 16$. Consequently the eight products $u^i v^{2j}$ are distinct, and $(i, j) \mapsto u^i v^{2j}$ identifies $\langle u, v^2 \rangle$ with $C_2 \times C_4$. The subgroup $\langle v \rangle$ is C_8 by the given order and the presentation. For the third subgroup,

$$(uv)^2 = u(vu)v = u(uv^5)v = v^6.$$

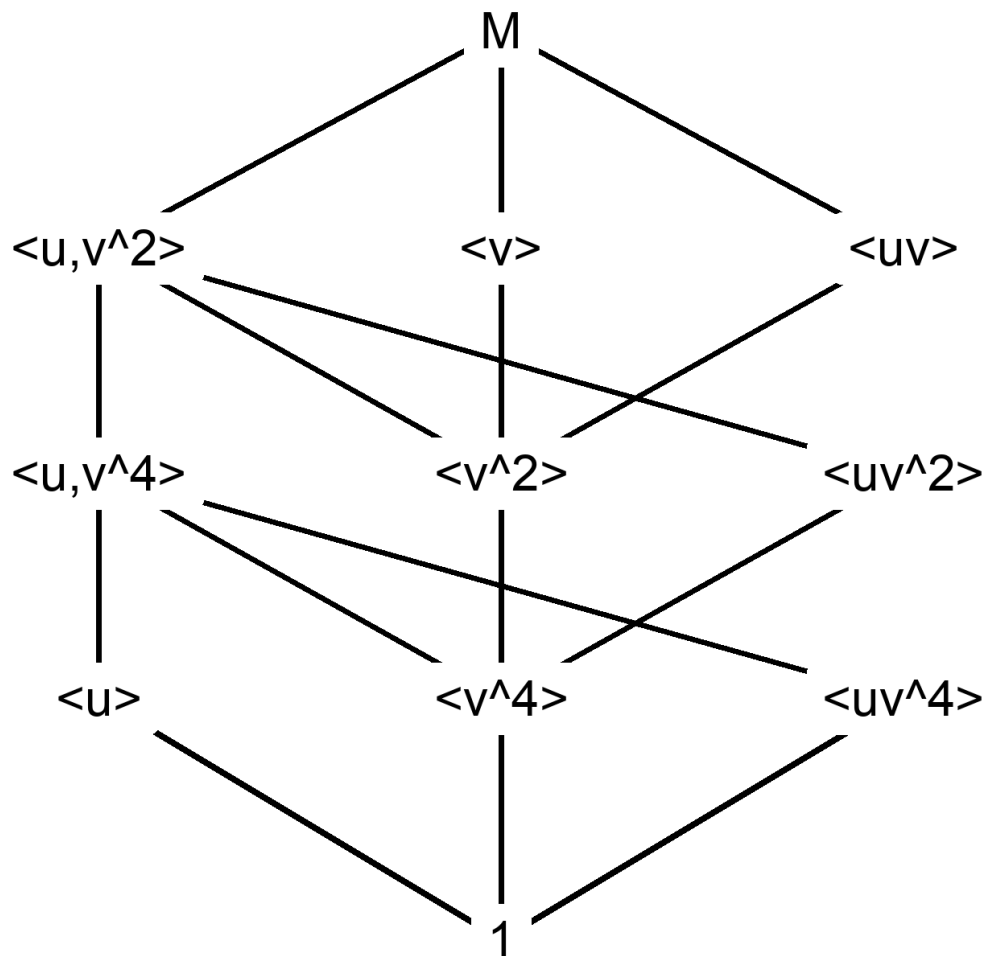
Since v^6 has order 4, uv has order 8: its eighth power is 1, while its fourth power is $v^{12} = v^4 \neq 1$. Thus $\langle uv \rangle \cong C_8$.

Step 3: glue the lattices. In $\langle u, v^2 \rangle$ the order-4 subgroups are $\langle u, v^4 \rangle$, $\langle v^2 \rangle$, $\langle uv^2 \rangle$ and the order-2 subgroups are $\langle u \rangle$, $\langle v^4 \rangle$, $\langle uv^4 \rangle$. Both cyclic maximal subgroups contain the chain

$$1 < \langle v^4 \rangle < \langle v^2 \rangle,$$

because $\langle (uv)^2 \rangle = \langle v^6 \rangle = \langle v^2 \rangle$. The problem says every proper subgroup is in a maximal subgroup on this list, so the list is complete. Substituting $x \mapsto u, y \mapsto v$ in the vertex labels of Exercise 13 preserves every cover relation. This is a lattice isomorphism, not a group homomorphism.

Step 4: distinguish the groups. The elements u, v do not commute: if $vu = uv$, comparison with $vu = uv^5$ would imply $v = v^5$ and hence $v^4 = 1$, contradicting $|v| = 8$. Thus M is nonabelian whereas $C_2 \times C_8$ is abelian. They cannot be isomorphic.



Technique: a subgroup lattice records inclusion, but it does not determine multiplication.

Foundations: Subgroup lattices and Hasse diagrams, Cosets, Lagrange, and multiplicativity of index.

3.1.11 — Upper triangular homomorphisms and fibers

Dummit–Foote, §3.1, Exercise 11; printed p. 86 (PDF p. 100). Statement normalized from the supplied book; solution independently derived.

Let F be a field and

$$G = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} : a, b, c \in F, ac \neq 0 \right\} \leq \mathrm{GL}_2(F).$$

- (a) Prove that $\varphi \left(\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \right) = a$ is a surjective homomorphism $G \rightarrow F^\times$. Describe its fibers and kernel.
- (b) Do the same for $\psi \left(\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \right) = (a, c)$ with codomain $F^\times \times F^\times$.
- (c) Prove that $H = \left\{ \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} : b \in F \right\}$ is isomorphic to the additive group of F .

Solution

For compactness write $T(a, b, c) = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$. Direct multiplication gives

$$T(a, b, c)T(a', b', c') = T(aa', ab' + bc', cc').$$

Since $a, c \neq 0$, inverses also exist in G :

$$T(a, b, c)^{-1} = T(a^{-1}, -a^{-1}bc^{-1}, c^{-1}).$$

These formulas also verify the subgroup assertion.

(a) The first diagonal entry. The product formula gives $\varphi(TT') = aa' = \varphi(T)\varphi(T')$. For any $t \in F^\times$, $T(t, 0, 1)$ maps to t , proving surjectivity. By definition a fiber consists of precisely those matrices with prescribed image. Therefore

$$\varphi^{-1}(t) = \{T(t, b, c) : b \in F, c \in F^\times\},$$

and

$$\ker \varphi = \varphi^{-1}(1) = \{T(1, b, c) : b \in F, c \in F^\times\}.$$

Do not impose $c = 1$ here: φ does not inspect the lower diagonal entry.

(b) Both diagonal entries. Multiplication in the codomain is coordinatewise, so $\psi(TT') = (aa', cc') = (a, c)(a', c')$. The matrix $T(t, 0, w)$ realizes any $(t, w) \in F^\times \times F^\times$. Thus ψ is surjective, and

$$\psi^{-1}(t, w) = \{T(t, b, w) : b \in F\}, \quad \ker \psi = \{T(1, b, 1) : b \in F\} = H.$$

(c) The additive field. Define $\theta : (F, +) \rightarrow H$ by $\theta(b) = T(1, b, 1)$. The multiplication formula reduces to $\theta(b)\theta(b') = T(1, b + b', 1) = \theta(b + b')$. Every element of H has this form, and equality of the upper-right entries shows that θ is injective. Hence θ is an isomorphism.

Technique: compute the matrix product first; then read off the image and solve the fiber equation without adding unintended restrictions.

Foundations: Normal subgroups, quotient groups, and coset order, Commutator criterion for abelian quotients.

3.1.22 — Arbitrary intersections of normal subgroups

Dummit–Foote, §3.1, Exercise 22; printed p. 88 (PDF p. 102). Statement normalized from the supplied book; solution independently derived.

(a) If $H, K \triangleleft G$, prove $H \cap K \triangleleft G$.

(b) Prove that the intersection of any nonempty family of normal subgroups of G is normal. The indexing set need not be countable.

Solution

(a) The identity belongs to both H and K . If $a, b \in H \cap K$, then ab^{-1} belongs to H and also to K , so $H \cap K$ is a subgroup by the one-step subgroup test. For $g \in G$ and $a \in H \cap K$, normality of each subgroup gives $gag^{-1} \in H$ and $gag^{-1} \in K$. Hence $g(H \cap K)g^{-1} \subseteq H \cap K$. Applying the same inclusion to g^{-1} yields the reverse inclusion after conjugating by g . Thus equality holds for every g .

(b) Let $(N_i)_{i \in I}$ be a family with $I \neq \emptyset$, and put $N = \bigcap_{i \in I} N_i$. We use the quantifier “for every $i \in I$,” not an enumeration. The identity lies in every N_i . If $a, b \in N$, then for every i , $a, b \in N_i$, so $ab^{-1} \in N_i$. Therefore $ab^{-1} \in N$. Similarly, if $a \in N$ and $g \in G$, then $gag^{-1} \in N_i$ for every i , so $gag^{-1} \in N$. The argument with g^{-1} again upgrades inclusion to equality. Hence $N \triangleleft G$.

Technique: prove a property in each member of the family, then put back the universal quantifier. No finiteness or countability is used.

Foundations: Normal subgroups, quotient groups, and coset order, Commutator criterion for abelian quotients.

3.1.28 — Checking a normalizer on generators

Dummit–Foote, §3.1, Exercise 28; printed p. 88 (PDF p. 102). Statement normalized from the supplied book; solution independently derived.

Let N be a finite subgroup of G and $N = \langle S \rangle$ for a subset S . Prove that $g \in G$ normalizes N if and only if $gSg^{-1} \subseteq N$.

Solution

Forward implication. If $g \in N_G(N)$, then $gNg^{-1} = N$. Since $S \subseteq N$, every gsg^{-1} belongs to N .

Reverse implication: first obtain inclusion. Suppose $gsg^{-1} \in N$ for every $s \in S$. Any element $n \in N$ is a finite word $s_1^{\epsilon_1} \cdots s_k^{\epsilon_k}$ with $s_i \in S$ and $\epsilon_i \in \{1, -1\}$. Conjugation preserves multiplication and inverses, so

$$gng^{-1} = (gs_1g^{-1})^{\epsilon_1} \cdots (gs_kg^{-1})^{\epsilon_k} \in N.$$

This also covers the identity, represented by the empty word. Thus $gNg^{-1} \subseteq N$.

Use the finite hypothesis. The map $n \mapsto gng^{-1}$ is a bijection $N \rightarrow gNg^{-1}$ with inverse conjugation by g^{-1} . Consequently $|gNg^{-1}| = |N|$. An inclusion between finite sets of equal cardinality is equality, so $gNg^{-1} = N$.

Why finiteness matters. In $\mathrm{GL}_2(\mathbb{Q})$ let $N = \left\{ \begin{pmatrix} 1 & m \\ 0 & 1 \end{pmatrix} : m \in \mathbb{Z} \right\}$, generated by the element with $m = 1$, and let $g = \mathrm{diag}(2, 1)$. Then conjugation sends m to $2m$. The generator is sent into N , but the conjugate subgroup contains only even upper-right entries and is proper in N .

Technique: “generators map into the subgroup” proves inclusion; a separate argument is needed for equality.

Foundations: Normal subgroups, quotient groups, and coset order, Commutator criterion for abelian quotients.

3.1.36 — A cyclic central quotient forces commutativity

Dummit–Foote, §3.1, Exercise 36; printed p. 89 (PDF p. 103). Statement normalized from the supplied book; solution independently derived.

Prove that if $G/Z(G)$ is cyclic, then G is abelian.

Solution

Let $xZ(G)$ generate $G/Z(G)$. This includes the trivial quotient: then we may take $x = 1$. Every $g \in G$ has $gZ(G) = x^a Z(G)$ for some integer a . Equality of these cosets means $x^{-a}g \in Z(G)$, so $g = x^a z$ for some $z \in Z(G)$.

Take arbitrary $g = x^a z$ and $h = x^b w$, with z, w central. Central elements commute with x and with one another, and powers of the same element commute. Therefore

$$gh = x^a z x^b w = x^{a+b} zw = x^{b+a} wz = x^b w x^a z = hg.$$

Since g, h were arbitrary, G is abelian.

Boundary check. An abelian quotient by an arbitrary normal subgroup does not imply G is abelian: $S_3/A_3 \cong C_2$. The argument works here because the kernel is the center, so the extra factors z, w commute with everything.

Technique: equality in a quotient gives a normal form modulo the kernel. Identify exactly what property of the kernel allows the next calculation.

Foundations: Normal subgroups, quotient groups, and coset order, Commutator criterion for abelian quotients.

3.1.41 — The commutator subgroup and abelianization

Dummit–Foote, §3.1, Exercise 41; printed p. 89 (PDF p. 103). Statement normalized from the supplied book; solution independently derived.

For a group G , let

$$N = \langle x^{-1}y^{-1}xy : x, y \in G \rangle.$$

Prove that $N \triangleleft G$ and G/N is abelian. Use the convention $[x, y] = x^{-1}y^{-1}xy$.

Solution

Step 1: normality of the generated subgroup. For $g, x, y \in G$, conjugation preserves products and inverses, so

$$g[x, y]g^{-1} = [gxg^{-1}, gyg^{-1}].$$

The right side is again one of the generators defining N . Conjugation by g therefore carries every word in those generators and their inverses into N . Thus $gNg^{-1} \subseteq N$. Applying the argument to g^{-1} gives equality. This proves normality without assuming N finite.

Step 2: commutativity in the quotient. Write $\bar{x} = xN$. Because $[x, y] \in N$,

$$\bar{x}^{-1}\bar{y}^{-1}\bar{x}\bar{y} = N.$$

Multiplying this equality on the left first by $\bar{x}$ and then by $\bar{y}$ gives $\bar{x}\bar{y} = \bar{y}\bar{x}$. Every quotient element is some xN , so the quotient is abelian.

A useful stronger conclusion. If $K \triangleleft G$ and G/K is abelian, the same quotient calculation shows $[x, y] \in K$ for every x, y . Therefore $N \leq K$. Conversely $N \leq K$ makes every commutator vanish in G/K . Thus

$$G/K \text{ abelian} \iff G' \leq K,$$

where $G' = N$. This criterion will be used in the solvability and perfect-group problems.

Technique: show that conjugation preserves a generating family, then turn a desired quotient identity into an element of the kernel.

Foundations: Normal subgroups, quotient groups, and coset order, Commutator criterion for abelian quotients.

PROBLEM • cu:algebra:midterm-2:section-3-2:pb:df-3-2-4

3.2.4 — Center of a group of order pq

Dummit–Foote, §3.2, Exercise 4; printed p. 95 (PDF p. 109). Statement normalized from the supplied book; solution independently derived.

If $|G| = pq$ for primes p, q , possibly equal, prove that either G is abelian or $Z(G) = 1$.

Solution

Suppose $Z(G) \neq 1$. By Lagrange, its order divides pq , so it is p, q , or pq (when $p = q$, the distinct possibilities are p, p^2).

If $|Z(G)| = pq$, then $Z(G) = G$, which is the definition of G being abelian. Otherwise $G/Z(G)$ has prime order: its order is q when $|Z(G)| = p$, or p when $|Z(G)| = q$. A group of prime order is cyclic, since any nonidentity element generates a nontrivial subgroup whose order divides that prime. Exercise 3.1.36 now gives that G is abelian.

Thus a nontrivial center forces G abelian. Equivalently, if G is not abelian, its center must be trivial.

Technique: constrain the order of the center, then inspect the central quotient. This alone does not assert that a nonabelian group of order pq exists.

Foundations: Cosets, Lagrange, and multiplicativity of index, First Isomorphism Theorem, Orbit–stabilizer and the kernel of an action.

3.2.8 — Coprime subgroup orders give trivial intersection

Dummit–Foote, §3.2, Exercise 8; printed p. 95 (PDF p. 109). Statement normalized from the supplied book; solution independently derived.

If H, K are finite subgroups of G with $\gcd(|H|, |K|) = 1$, prove $H \cap K = 1$.

Solution

The intersection is a subgroup of both H and K . By Lagrange, $|H \cap K|$ divides $|H|$ and also $|K|$. The only positive common divisor is 1 , so $|H \cap K| = 1$. A subgroup of order 1 consists only of the identity, proving the result.

Check the converse. Trivial intersection does not force coprime orders: distinct order- 2 subgroups of $C_2 \times C_2$ intersect trivially. The statement is a sufficient criterion, not an equivalence.

Technique: the same subgroup supplies a divisor of two different orders.

Foundations: Cosets, Lagrange, and multiplicativity of index, First Isomorphism Theorem, Orbit–stabilizer and the kernel of an action.

3.2.9 — Cauchy's theorem by cyclic rotation — (a)–(f)

Dummit–Foote, §3.2, Exercise 9; printed p. 96 (PDF p. 110). Statement normalized from the supplied book; solution independently derived.

Let G be finite and let the prime p divide $|G|$. Define

$$S = \{(x_1, \dots, x_p) \in G^p : x_1 \cdots x_p = 1\}.$$

Relate two tuples when one is a cyclic rotation of the other.

(a) Show $|S| = |G|^{p-1}$ and $p \mid |S|$.

(b) Show rotations preserve S .

(c) Show that this relation is an equivalence relation.

(d) Show singleton classes are exactly the constant tuples $(x, \dots, x)$ with $x^p = 1$.

(e) Show every class has size 1 or p . If there are k singleton classes and d classes of size p , deduce $|G|^{p-1} = k + pd$.

(f) Deduce there is a nonidentity element of order p .

Solution

Let $r(x_1, \dots, x_p) = (x_2, \dots, x_p, x_1)$.

(a) Count by free coordinates. Choose $x_1, \dots, x_{p-1}$ arbitrarily. There is exactly one possible last coordinate,

$$x_p = (x_1 \cdots x_{p-1})^{-1},$$

which makes the product 1. This gives a bijection $G^{p-1} \rightarrow S$ and hence $|S| = |G|^{p-1}$. Since a prime is at least 2, $p - 1 \geq 1$, and $p \mid |G|$ implies $p \mid |S|$.

(b) Preserve the product in the noncommutative case. If $x_1 \cdots x_p = 1$, then

$$x_2 \cdots x_p x_1 = x_1^{-1} (x_1 \cdots x_p) x_1 = 1.$$

So $r(S) \subseteq S$. Repeating this proves every cyclic rotation lies in S . We did not commute any of the factors.

(c) Verify the equivalence relation. The relation says $\beta = r^j \alpha$ for an integer j . Taking $j = 0$ proves reflexivity. If $\beta = r^j \alpha$, then $\alpha = r^{-j} \beta$, proving symmetry; negative powers are legitimate because $r^p = \text{id}$. If also $\gamma = r^\ell \beta$, then $\gamma = r^{\ell+j} \alpha$, proving transitivity.

(d) Characterize singleton classes. A singleton class means $r\alpha = \alpha$. Coordinate comparison gives $x_1 = x_2 = \dots = x_p = x$. Membership in S then says $x^p = 1$. Conversely such a constant tuple is unchanged by all rotations and belongs to S , so its class is a singleton.

(e) Explain why no intermediate sizes occur. The group $C_p = \langle r \rangle$ acts on S . By orbit-stabilizer, the size of a class is the index of its stabilizer in C_p and therefore divides p . Since p is prime, that positive size is 1 or p . One may also argue directly: if a nonzero rotation r^j fixes a tuple, $1 \leq j < p$, then j has a multiplicative inverse modulo p , so a power of r^j is r and r fixes the tuple. Hence any nonsingleton has all p distinct rotations. Counting the partition into classes gives $|G|^{p-1} = k + pd$.

(f) Find an element of order p . Part (a) and the equation in (e) imply $p \mid k$. The tuple $(1, \dots, 1)$ provides one singleton, so $k \geq 1$. A positive multiple of p is at least p , thus $k \geq p > 1$. There must be another singleton $(x, \dots, x)$ with $x \neq 1$ and $x^p = 1$. The order of x divides p , and cannot be 1, so it is p .

Technique: count a finite set modulo p in two ways. Prime-order symmetry isolates fixed points. The rotation calculation is the step where an unjustified commutativity assumption most often sneaks in.

Foundations: Cosets, Lagrange, and multiplicativity of index, First Isomorphism Theorem, Orbit-stabilizer and the kernel of an action.

3.2.18 — Coprime order and quotient index

Dummit–Foote, §3.2, Exercise 18; printed p. 96 (PDF p. 110). Statement normalized from the supplied book; solution independently derived.

Let G be finite, $H \leq G$, and $N \triangleleft G$. If $\gcd(|H|, [G : N]) = 1$, prove $H \leq N$.

Solution

Let $\pi : G \rightarrow G/N$ be the natural projection and restrict it to H . The restriction is a homomorphism with kernel $H \cap N$. By the First Isomorphism Theorem,

$$\pi(H) \cong H/(H \cap N),$$

so $|\pi(H)|$ divides $|H|$. On the other hand $\pi(H)$ is a subgroup of G/N , so its order divides $|G/N| = [G : N]$. Coprimality forces $|\pi(H)| = 1$. Thus $\pi(h) = N$ for every $h \in H$, which means $h \in N$. Therefore $H \leq N$.

Technique: look at the image in the quotient. Its order is simultaneously constrained by the domain and the codomain. Normality is needed to make G/N a group.

Foundations: Cosets, Lagrange, and multiplicativity of index, First Isomorphism Theorem, Orbit–stabilizer and the kernel of an action.

Additional problem — Minimal abelian normal subgroup and complement

Additional problem supplied directly in the request; solution independently derived.

Suppose N is a nontrivial abelian subgroup of G , minimal among nontrivial normal subgroups of G contained in N . Let $H < G$ satisfy $NH = G$. Prove $N \cap H = 1$ and H is a maximal subgroup of G .

Here the minimality wording in the supplied problem is read as: $N \triangleleft G$ and no subgroup $1 < L < N$ is normal in G . Since N is abelian, all its subgroups are abelian.

Solution

Step 1: show the intersection is normal in G . Put $L = N \cap H$. For $h \in H$, normality of N gives $hNh^{-1} = N$, and $hHh^{-1} = H$ because h belongs to H . Hence $hLh^{-1} = L$. Every $n \in N$ centralizes L , since $L \leq N$ and N is abelian; thus $nLn^{-1} = L$. Every $g \in G$ is nh because $NH = G$, so

$$gLg^{-1} = n(hLh^{-1})n^{-1} = nLn^{-1} = L.$$

Therefore $L \triangleleft G$.

Step 2: apply minimality and properness. The subgroup L lies in N , so it is either 1 or N . If $L = N$, then $N \leq H$, which implies $G = NH = H$, contrary to $H < G$. Consequently $N \cap H = 1$.

Step 3: test an arbitrary overgroup. Let $H < K \leq G$. Choose $k \in K \setminus H$, and write $k = nh$ with $n \in N, h \in H$. Then $n = kh^{-1} \in K \cap N$. Also $n \neq 1$, since $n = 1$ would give $k = h \in H$. Thus $N \cap K \neq 1$.

As in Step 1, $N \cap K$ is normalized by H : elements of H normalize K because $H \leq K$, and normalize N because $N \triangleleft G$. It is centralized by N because N is abelian. Since $G = NH$, this proves $N \cap K \triangleleft G$. Minimality therefore gives $N \cap K = N$, so $N \leq K$. We already have $H \leq K$, hence $G = NH \leq K$. Thus $K = G$.

Every subgroup strictly larger than H is therefore G , which is exactly maximality. No finite-order hypothesis is needed.

Technique: turn an intersection into a normal subgroup using the two factors N, H ; minimality gives a dichotomy. Then repeat that argument for each possible overgroup.

Foundations: Cosets, Lagrange, and multiplicativity of index, First Isomorphism Theorem, Orbit–stabilizer and the kernel of an action.

3.3.6 — Quotient lattice of the modular group

Dummit–Foote, §3.3, Exercise 6; printed p. 101 (PDF p. 115). Statement normalized from the supplied book; solution independently derived.

Let $M = \langle u, v \mid u^2 = v^8 = 1, vu = uv^5 \rangle$ be the modular group of order 16. Prove $\langle v^4 \rangle \triangleleft M$. Use the Lattice Isomorphism Theorem to draw the subgroup lattice of $M/\langle v^4 \rangle$, compare it with groups of order 8, and determine its isomorphism type using generators and relations.

Solution

Step 1: normality. Conjugation by v fixes v^4 . Conjugation by u gives $uv^4u = (v^5)^4 = v^{20} = v^4$. The generators u, v thus centralize v^4 , so $\langle v^4 \rangle \leq Z(M)$ and is normal.

Step 2: lift the quotient vertices. Write $L = \langle v^4 \rangle$. It has order 2, so the quotient has order 8. By the lattice correspondence, its subgroups are exactly T/L for subgroups $T \leq M$ containing L . From the complete lattice in 2.5.14 these are:

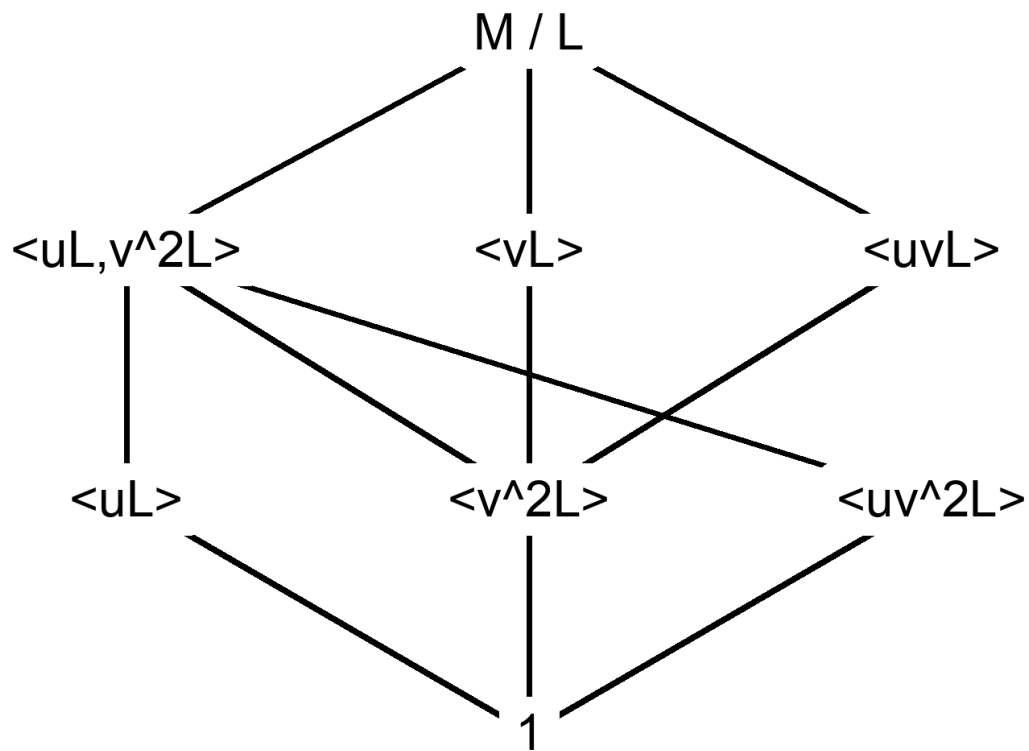
- L itself, giving the identity vertex;
- $\langle u, v^4 \rangle, \langle v^2 \rangle, \langle uv^2 \rangle$, giving three order-2 vertices;
- $\langle u, v^2 \rangle, \langle v \rangle, \langle uv \rangle$, giving three order-4 vertices;
- M , giving the top vertex.

Let bars denote cosets modulo L . The order-4 vertices are $\langle \bar{u}, \bar{v}^2 \rangle, \langle \bar{v} \rangle, \langle \bar{u}\bar{v} \rangle$. The first covers all three order-2 vertices $\langle \bar{u} \rangle, \langle \bar{v}^2 \rangle, \langle \bar{u}\bar{v}^2 \rangle$; each of the other two covers only $\langle \bar{v}^2 \rangle$. All order-2 vertices cover 1, and the whole quotient covers the three order-4 vertices. The diagram below is exactly the lattice of $C_2 \times C_4$.

Step 3: establish the group type, not just the lattice. In the quotient $\bar{v}^4 = 1$, and the defining relation becomes $\bar{v}\bar{u} = \bar{u}\bar{v}^5 = \bar{u}\bar{v}$. Thus the two quotient generators commute. Also $\bar{u}^2 = 1$. Define

$$f : C_2 \times C_4 \longrightarrow M/L, \quad f(i, j) = \bar{u}^i \bar{v}^j.$$

The order relations make this well defined, and commutativity makes it a homomorphism. It is surjective because $\bar{u}, \bar{v}$ generate the quotient. Domain and codomain both have order 8, so it is bijective. Hence $M/L \cong C_2 \times C_4$.



Technique: correspondence gives the lattice; a surjective map plus a cardinality check determines the actual group.

Foundations: First Isomorphism Theorem, Second (diamond) Isomorphism Theorem, Subgroup correspondence (Lattice Isomorphism Theorem), Products of subgroups and their orders.

3.3.7 — Two normal factors and a product quotient

Dummit–Foote, §3.3, Exercise 7; printed p. 101 (PDF p. 115). Statement normalized from the supplied book; solution independently derived.

Let $M, N \triangleleft G$ and $G = MN$. Prove

$$G/(M \cap N) \cong (G/M) \times (G/N).$$

Include the subgroup relationships used in the argument.

Solution

Define

$$f : G \rightarrow (G/M) \times (G/N), \quad f(g) = (gM, gN).$$

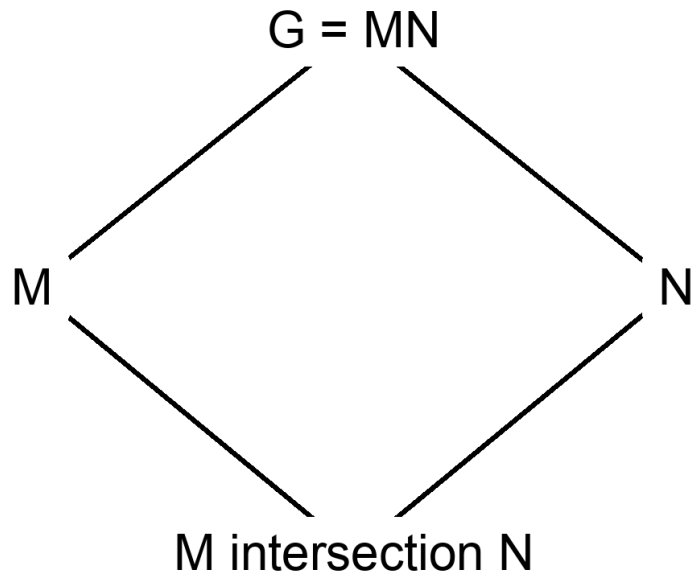
Normality makes both quotient groups legitimate and gives $f(gh) = (ghM, ghN) = f(g)f(h)$. The kernel is exactly $M \cap N$, since both coordinates are identity cosets precisely when g belongs to both subgroups.

The point requiring care is surjectivity. Take arbitrary target cosets (aM, bN) . Since $G = MN$, write $a^{-1}b = mn$ with $m \in M, n \in N$. Set $g = am$. Then $gM = aM$ because $m \in M$. Also $b = amn = gn$, so $gN = bN$. Thus every pair of cosets has a preimage.

The First Isomorphism Theorem now gives the stated isomorphism, with induced map $g(M \cap N) \mapsto (gM, gN)$.

The relevant lattice is the diamond with top $G = MN$, middle vertices M, N , and bottom $M \cap N$. The lines here express inclusion and need not be covers. The second isomorphism theorem identifies $M/(M \cap N)$ with G/N and $N/(M \cap N)$ with G/M .

Check the assumption. Without $G = MN$, the map still exists and has the same kernel, but its image need not be the full product. For $M = N = 1$ it is the diagonal subgroup of $G \times G$.



Inclusion diagram; lines need not be covers

Technique: use a single map to the product. Identify its kernel and prove joint surjectivity rather than just surjectivity of each coordinate.

Foundations: First Isomorphism Theorem, Second (diamond) Isomorphism Theorem, Subgroup correspondence (Lattice Isomorphism Theorem), Products of subgroups and their orders.

3.3.9 — Intersecting a Sylow subgroup with a normal subgroup

Dummit–Foote, §3.3, Exercise 9; printed p. 101 (PDF p. 115). Statement normalized from the supplied book; solution independently derived.

Let $|G| = p^a m$ with $p \nmid m$, let $P \leq G$ have order p^a , and let $N \triangleleft G$ have order $p^b n$ with $p \nmid n$. Prove $|P \cap N| = p^b$ and $|PN/N| = p^{a-b}$.

Solution

Because $P \cap N \leq P$, its order is a power p^c . Since it is also a subgroup of N , Lagrange gives $c \leq b$. Normality of N makes PN a subgroup, and the product-order formula gives

$$|PN| = \frac{|P||N|}{|P \cap N|} = p^{a+b-c}n.$$

This subgroup order divides $|G| = p^a m$. Since $p \nmid n, m$, the exponent of p in its order cannot exceed a . Hence $a + b - c \leq a$, or $b \leq c$. Together with $c \leq b$, this forces $c = b$.

Therefore $|P \cap N| = p^b$. The second isomorphism theorem gives

$$PN/N \cong P/(P \cap N),$$

so its order is $p^a/p^b = p^{a-b}$. In particular $P \cap N$ has the largest possible p -power order in N , and PN/N has the largest possible p -power order in G/N .

Technique: introduce an unknown exponent for the intersection; Lagrange bounds it in one direction and the product formula in the other. No Sylow existence theorem is needed because P is already supplied.

Foundations: First Isomorphism Theorem, Second (diamond) Isomorphism Theorem, Subgroup correspondence (Lattice Isomorphism Theorem), Products of subgroups and their orders.

3.3.10 — Hall subgroups under intersection and quotient

Dummit–Foote, §3.3, Exercise 10; printed p. 101 (PDF p. 115). Statement normalized from the supplied book; solution independently derived.

A subgroup H of a finite group G is Hall if $\gcd(|H|, [G : H]) = 1$. If H is Hall and $N \triangleleft G$, prove $H \cap N$ is Hall in N and HN/N is Hall in G/N .

Solution

Put $h = |H|$, $d = [G : H]$, and $t = |H \cap N|$. Since HN is a subgroup containing H , multiplicativity of indices gives

$$[G : H] = [G : HN][HN : H].$$

The product-order formula gives $[HN : H] = |N|/t = [N : H \cap N]$. Thus $[N : H \cap N]$ divides d . Also t divides h . A common prime divisor of t and $[N : H \cap N]$ would then divide both h and d , contradicting the Hall condition. Hence $H \cap N$ is Hall in N .

Next, the second isomorphism theorem gives $|HN/N| = h/t$, which divides h . The index of that subgroup in G/N is

$$[G/N : HN/N] = [G : HN],$$

which divides d by the first displayed equality. Again a common prime divisor would divide both h and d . Therefore HN/N is Hall in G/N .

Technique: prove the two quantities whose gcd is needed divide the original coprime pair. This avoids assuming H itself is normal.

Foundations: First Isomorphism Theorem, Second (diamond) Isomorphism Theorem, Subgroup correspondence (Lattice Isomorphism Theorem), Products of subgroups and their orders.

3.4.2 — All composition series of Q_8 and D_8

Dummit–Foote, §3.4, Exercise 2; printed p. 106 (PDF p. 120). Statement normalized from the supplied book; solution independently derived.

Exhibit all three composition series of Q_8 and all seven composition series of D_8 . List the composition factors for each series. Use $D_8 = \langle r, s \mid r^4 = s^2 = 1, srs = r^{-1} \rangle$.

Solution

A composition series is a chain from 1 to the group in which each term is normal in the next and every successive quotient is nontrivial simple. A group of order 2 is simple, so every normal step of index 2 is already a composition step.

Q_8 . Its unique order- 2 subgroup is $Z = \langle -1 \rangle$. Its three order- 4 subgroups are $\langle i \rangle, \langle j \rangle, \langle k \rangle$, each cyclic. The three series are

$$1 < Z < \langle i \rangle < Q_8,$$

$$1 < Z < \langle j \rangle < Q_8,$$

$$1 < Z < \langle k \rangle < Q_8.$$

In each chain all steps have index 2 and hence are normal. Every factor is C_2 .

D_8 . Let $R = \langle r \rangle$, $A = \langle r^2, s \rangle$, and $B = \langle r^2, sr \rangle$. These are the three order- 4 subgroups; $R \cong C_4$ and $A, B \cong V_4$. The seven series are

$$1 < \langle r^2 \rangle < R < D_8,$$

$$1 < \langle r^2 \rangle < A < D_8,$$

$$1 < \langle s \rangle < A < D_8,$$

$$1 < \langle sr^2 \rangle < A < D_8,$$

$$1 < \langle r^2 \rangle < B < D_8,$$

$$1 < \langle sr \rangle < B < D_8,$$

$$1 < \langle sr^3 \rangle < B < D_8.$$

Again all factors in every series are C_2 , in the order C_2, C_2, C_2 from bottom to top. The order-2 term need only be normal in its immediate order-4 overgroup; reflection subgroups need not be normal in all of D_8 .

Why these are all. In either group, a composition factor has order 2, 4, or 8, since its order divides 8. No group of order 4 is simple: Cauchy's theorem gives a subgroup of order 2, which has index 2 and is normal. Neither Q_8 nor D_8 is simple, since each has proper normal subgroups of index 2. Thus no composition step can skip from order 1 to 4, from 2 to 8, or from 1 to 8. The series must pass through orders 1, 2, 4, 8. The lists above exhaust all choices at those orders.

Technique: enumerate maximal normal subgroups, then work down inside each one. Normality is required between adjacent terms, not necessarily in the whole group.

Foundations: Simple groups and composition series, Existence and uniqueness of composition factors, Derived sequence criterion and solvable extensions.

3.4.7 — A composition series through a given normal subgroup

Dummit–Foote, §3.4, Exercise 7; printed p. 106 (PDF p. 120). Statement normalized from the supplied book; solution independently derived.

Let G be finite and $H \triangleleft G$. Prove there is a composition series of G having H as one of its terms.

Solution

Step 1: take series on the two sides of H . Every finite group has a composition series: if it is nontrivial, select a maximal proper normal subgroup, whose quotient is simple, and continue inside that subgroup. Strictly decreasing finite orders guarantee termination. Apply this construction to H and to G/H .

Write a composition series of H as

$$1 = H_0 \triangleleft H_1 \triangleleft \dots \triangleleft H_t = H.$$

Write one of G/H as

$$1 = \bar{K}_0 \triangleleft \bar{K}_1 \triangleleft \dots \triangleleft \bar{K}_s = G/H.$$

Step 2: pull the quotient series back. Let $\pi : G \rightarrow G/H$ be projection and define $K_i = \pi^{-1}(\bar{K}_i)$. Then $K_0 = H$, $K_s = G$, and $K_i \leq K_{i+1}$. To check adjacent normality, take $k \in K_{i+1}$ and $a \in K_i$. Their images satisfy $\pi(k)\pi(a)\pi(k)^{-1} \in \bar{K}_i$, so $kak^{-1} \in K_i$. Thus $K_i \triangleleft K_{i+1}$.

The map $K_{i+1} \rightarrow \bar{K}_{i+1}/\bar{K}_i$, sending k to $\pi(k)\bar{K}_i$, is surjective and has kernel K_i . Hence

$$K_{i+1}/K_i \cong \bar{K}_{i+1}/\bar{K}_i,$$

which is simple. Every pulled-back step is therefore a composition step.

Step 3: concatenate. The chain

$$1 = H_0 \triangleleft \dots \triangleleft H_t = H = K_0 \triangleleft \dots \triangleleft K_s = G$$

is the desired series. Count the shared term $H = K_0$ only once. If $H = 1$ or $H = G$, the corresponding series is empty on that side, and the argument still works.

Technique: construct inside the normal subgroup and inside the quotient, then lift and concatenate.

Foundations: Simple groups and composition series, Existence and uniqueness of composition factors, Derived sequence criterion and solvable extensions.

3.4.11 — An abelian normal subgroup inside a solvable group

Dummit–Foote, §3.4, Exercise 11; printed p. 106 (PDF p. 120). Statement normalized from the supplied book; solution independently derived.

Let G be solvable and $1 \neq H \triangleleft G$. Prove there is a nontrivial abelian subgroup $A \leq H$ with $A \triangleleft G$.

Hint supplied in the request: prove that if $N \triangleleft G$, then $[N, N] \triangleleft G$.

Solution

Step 1: verify the hint. Use $[x, y] = x^{-1}y^{-1}xy$ and let $N' = [N, N]$ be the subgroup generated by commutators of elements of N . For $g \in G$ and $x, y \in N$,

$$g[x, y]g^{-1} = [gxg^{-1}, gyg^{-1}].$$

Since $N \triangleleft G$, the two conjugated elements still lie in N . Conjugation therefore sends each generator of N' into N' and sends every word in them into N' . The same statement for g^{-1} proves equality, so $N' \triangleleft G$.

Step 2: form the derived sequence inside H . Put $H^{(0)} = H$ and $H^{(j+1)} = [H^{(j)}, H^{(j)}]$. Step 1 proves inductively that each $H^{(j)}$ is normal in G .

We need to know this sequence reaches 1 . The lecture proves that the derived sequence of a solvable group reaches 1 . Here is the direct reason: if $G = G_0 \supseteq G_1 \supseteq \dots \supseteq G_t = 1$ has abelian successive quotients, then $G'_i \leq G_{i+1}$. Inductively $G^{(i)} \leq G_i$, because commutators of a subgroup are commutators in its overgroup. Hence $G^{(t)} = 1$. Since $H \leq G$, induction similarly gives $H^{(j)} \leq G^{(j)}$, so $H^{(t)} = 1$ too.

Step 3: choose the last nontrivial term. Since $H \neq 1$, there is a last index j with $H^{(j)} \neq 1$. Let $A = H^{(j)}$. Then $A \leq H$ and $A \triangleleft G$ by Step 2. Its commutator subgroup is $A' = H^{(j+1)} = 1$. Thus every commutator of two elements of A is the identity, which is equivalent to those elements commuting. Therefore A is abelian and nontrivial, as required.

Technique: take the last nonzero term of a terminating sequence. The hint preserves normality while the derived sequence strips away noncommutativity. This proof does not require G finite.

Foundations: Simple groups and composition series, Existence and uniqueness of composition factors, Derived sequence criterion and solvable extensions.

3.5.4 — Two generators for S_n

Dummit–Foote, §3.5, Exercise 4; printed p. 111 (PDF p. 125). Statement normalized from the supplied book; solution independently derived.

Prove $S_n = \langle (1\ 2), (1\ 2\ \dots\ n) \rangle$ for every $n \geq 2$.

Solution

Let $t = (1\ 2)$, $c = (1\ 2\ \dots\ n)$, and $L = \langle t, c \rangle$. Conjugating a transposition just relabels its two points, so for $0 \leq j \leq n-2$,

$$c^j t c^{-j} = (j+1\ j+2).$$

Thus L contains every adjacent transposition.

Why adjacent transpositions suffice. Let $s_i = (i\ i+1)$. For $i < j$, the permutation $w = s_i s_{i+1} \dots s_{j-2}$ fixes j and sends $j-1$ to i (products act right to left). Hence

$$w s_{j-1} w^{-1} = (i\ j).$$

So every transposition belongs to the subgroup generated by adjacent transpositions. Every cycle is a product of transpositions:

$$(a_1\ a_2\ \dots\ a_k) = (a_1\ a_k)(a_1\ a_{k-1}) \dots (a_1\ a_2).$$

Disjoint-cycle decomposition therefore writes every permutation as a product of transpositions. Consequently L contains all of S_n . Since $L \leq S_n$ by construction, equality follows. For $n = 2$, $t = c$ already generates S_2 .

Technique: conjugate one simple generator to obtain a familiar generating family, then explicitly show that family generates.

Foundations: Technique — Conjugate supports and pair transpositions, Simple groups and composition series.

3.5.6 — A proper dihedral subgroup of S_4

Dummit–Foote, §3.5, Exercise 6; printed p. 111 (PDF p. 125). Statement normalized from the supplied book; solution independently derived.

Show that $\langle (1\ 3), (1\ 2\ 3\ 4) \rangle$ is proper in S_4 and determine its isomorphism type.

Solution

Put $s = (1\ 3)$ and $r = (1\ 2\ 3\ 4)$. Then $s^2 = 1$, $r^4 = 1$, and conjugating r by s gives

$$srs = (3\ 2\ 1\ 4) = (1\ 4\ 3\ 2) = r^{-1}.$$

Using $sr = r^{-1}s$, every word in r, s can be rewritten as r^i or sr^i , with $0 \leq i < 4$. Thus the generated group has at most eight elements.

The four powers of r are distinct because r has order 4. Also $s \notin \langle r \rangle$: the nonidentity powers of r are two 4-cycles and the double transposition $(1\ 3)(2\ 4)$, none of which is s . Hence the four elements sr^i are distinct from the powers of r and from one another. There are exactly eight elements.

The standard dihedral presentation maps onto this group by sending its rotation and reflection to r, s . Both groups have order 8, so the map is an isomorphism. Thus the subgroup is D_8 . Since $|S_4| = 24$, it is proper.

Technique: relations give an upper bound by normal forms; distinguish the normal forms to obtain the matching lower bound.

Foundations: Technique — Conjugate supports and pair transpositions, Simple groups and composition series.

3.5.10 — A composition series for A_4

Dummit–Foote, §3.5, Exercise 10; printed p. 111 (PDF p. 125). Statement normalized from the supplied book; solution independently derived.

Find a composition series for A_4 and deduce that A_4 is solvable.

Solution

Define

$$V = \{1, (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\}.$$

The three nonidentity elements have order 2, and the product of any two distinct ones is the third. Therefore V is a subgroup isomorphic to V_4 . Conjugation in S_4 relabels the points of a double transposition, so it preserves this set. Hence $V \triangleleft A_4$.

Take $L = \langle (1\ 2)(3\ 4) \rangle$. The chain

$$1 \triangleleft L \triangleleft V \triangleleft A_4$$

has factors $L \cong C_2$, $V/L \cong C_2$, and $A_4/V \cong C_3$, since their orders are 2, 2, 3. Each is simple because it has prime order, so the chain is a composition series. Each is also abelian, so this very chain satisfies the definition of solvability.

Careful distinction. The subgroup L is normal in V , but not in A_4 : a 3-cycle conjugates its nonidentity element to a different double transposition. That does not invalidate the composition or solvable series; adjacent normality is the required condition.

Technique: find a small normal subgroup recognizable from cycle type, then refine it internally.

Foundations: Technique — Conjugate supports and pair transpositions, Simple groups and composition series.

4.1.7 — Blocks, primitivity, and maximal point stabilizers

Dummit–Foote, §4.1, Exercise 7; printed p. 117 (PDF p. 131). Statement normalized from the supplied book; solution independently derived.

Let G be a transitive permutation group on a finite nonempty set A . A block is a nonempty $B \subseteq A$ such that for every $g \in G$, either $gB = B$ or $gB \cap B = \emptyset$.

- (a) If $a \in B$, prove the setwise stabilizer $G_B = \{g : gB = B\}$ is a subgroup containing the point stabilizer G_a .
- (b) Show the distinct translates of B partition A .
- (c) A transitive action is primitive if only singletons and A are blocks. Show S_4 is primitive in its natural action, whereas D_8 on a square's vertices is not.
- (d) Prove primitivity is equivalent to each point stabilizer being maximal in G .

Solution

(a) Setwise stabilization. The identity stabilizes B . If $gB = B$ and $hB = B$, then $h^{-1}B = B$ and $gh^{-1}B = B$, proving $G_B \leq G$. If $g \in G_a$, then $a = g(a) \in gB \cap B$. Since B is a block, this nonempty intersection forces $gB = B$. Hence $G_a \leq G_B$.

(b) The translates form a partition. If $gB \cap hB \neq \emptyset$, apply h^{-1} to get $(h^{-1}g)B \cap B \neq \emptyset$. The block condition yields $(h^{-1}g)B = B$, so $gB = hB$. Thus distinct translates are disjoint. They cover A because, fixing $a \in B$, transitivity supplies for each $b \in A$ some g with $g(a) = b$, whence $b \in gB$.

(c) The two examples. Suppose B is a block for S_4 with $1 < |B| < 4$. Choose distinct $a, b \in B$ and $c \notin B$. The transposition $(b\ c)$ fixes a , so its translate of B intersects B , but the translate contains c and is not B . This contradicts the block condition. Hence only trivial blocks exist. For D_8 , label square vertices 1, 2, 3, 4 cyclically. The opposite pair $\{1, 3\}$ is sent by every square symmetry either to itself or to the disjoint opposite pair $\{2, 4\}$. It is a nontrivial block.

(d) Maximal stabilizers imply primitivity. Assume $|A| > 1$ and G_a maximal. For a block B containing a , part (a) gives $G_a \leq G_B \leq G$, so $G_B = G_a$ or $G_B = G$. If $G_B = G$, transitivity and invariance of B imply $B = A$. If $G_B = G_a$, let $b \in B$ and choose g with $g(a) = b$. Then gB intersects B at b , so $g \in G_B = G_a$ and $b = a$. Thus $B = \{a\}$. Any block can be treated by choosing a point it contains.

Primitivity implies maximal stabilizers. Let $G_a \leq K \leq G$ and set $B = K \cdot a$. To prove B is a block, suppose $gB \cap B \neq \emptyset$. Then $gk_1(a) = k_2(a)$ for some $k_1, k_2 \in K$. Thus $k_2^{-1}gk_1 \in G_a \leq K$, implying $g \in K$ and hence $gB = B$. Primitivity forces $B = \{a\}$ or A . In the first case every $k \in K$ fixes a , so $K = G_a$. In the second case, for arbitrary $g \in G$, choose $k \in K$ with $k(a) = g(a)$. Then $k^{-1}g \in G_a \leq K$, so $g \in K$. Hence $K = G$. For $|A| > 1$, transitivity makes $G_a < G$, so this is maximality.

Boundary in the textbook wording. The equivalence in (d) needs $|A| > 1$. On a singleton the action is primitive according to the definition, but $G_a = G$ is not a proper, hence not a maximal, subgroup. Parts (a)–(c) have no such problem. The proof above states the missing boundary condition explicitly.

Technique: an intersecting translate forces a block to be invariant. Overgroups of a stabilizer correspond to blocks through its point.

Foundations: Orbit–stabilizer and the kernel of an action, Actions, orbits, stabilizers, and kernels.

4.1.9 — Orbits of a normal subgroup

Dummit–Foote, §4.1, Exercise 9; printed p. 117 (PDF p. 131). Statement normalized from the supplied book; solution independently derived.

Suppose G acts transitively on a finite nonempty set A , and $H \triangleleft G$. Let $O_1, \dots, O_r$ be the H -orbits.

(a) Prove G permutes these orbits transitively, and all have equal cardinality.

(b) For $a \in O_1$, prove $|O_1| = [H : H \cap G_a]$ and $r = [G : HG_a]$.

Solution

(a) A translate is another H -orbit. For $a \in A$ and $g \in G$,

$$g(H \cdot a) = \{(ghg^{-1}) \cdot (g \cdot a) : h \in H\} = H \cdot (g \cdot a),$$

because normality says $gHg^{-1} = H$. Thus g sends an entire H -orbit onto an H -orbit. This defines a G -action on the collection of orbits. Given O_i, O_j , choose $a_i \in O_i, a_j \in O_j$. Transitivity on A gives $g(a_i) = a_j$, and the displayed equality then gives $gO_i = O_j$. Hence the action on the orbit collection is transitive. Since g is a bijection of A , its restriction is a bijection $O_i \rightarrow O_j$, proving equal sizes. Distinct H -orbits are disjoint, so each is also a block for G .

(b) Count within an orbit. The stabilizer of a in H is $H_a = H \cap G_a$. Orbit–stabilizer for the restricted action of H gives $|O_1| = [H : H \cap G_a]$.

Count the orbit collection. Its stabilizer at $O_1 = H \cdot a$ consists exactly of HG_a . If $gO_1 = O_1$, then $g(a) = h(a)$ for some $h \in H$, so $h^{-1}g \in G_a$ and $g \in HG_a$. Conversely if $g = hk$ with $k \in G_a$, then $g(a) = h(a) \in O_1$, and part (a) says gO_1 is the H -orbit containing this point, hence O_1 . As H is normal, HG_a is a subgroup. Orbit–stabilizer for the transitive action on the r orbits therefore yields $r = [G : HG_a]$.

Technique: apply orbit–stabilizer twice: once to points under H , once to H -orbits under G . Identify the latter stabilizer explicitly.

Foundations: Orbit–stabilizer and the kernel of an action, Actions, orbits, stabilizers, and kernels.

4.2.9 — Index- p subgroups in p -groups

Dummit–Foote, §4.2, Exercise 9; printed p. 122 (PDF p. 136). Statement normalized from the supplied book; solution independently derived.

Let p be prime and $|G| = p^a$ with $a \geq 1$. Prove every subgroup of index p is normal. Deduce every group of order p^2 has a normal subgroup of order p .

Solution

Let $H \leq G$ have index p . Left multiplication on the p left cosets defines a homomorphism $\varphi : G \rightarrow S_p$. Let $K = \ker \varphi$. An element of K fixes the coset H , so it belongs to H ; hence $K \leq H$.

By the First Isomorphism Theorem the image has order $[G : K]$, which is a power of p because $|G| = p^a$. It divides $p!$ because it is a subgroup of S_p . There is only one factor divisible by p in the list $1, 2, \dots, p$, so the highest power of p dividing $p!$ is p itself. Thus $[G : K] \leq p$. But

$$[G : K] = [G : H][H : K] = p[H : K] \geq p.$$

Therefore $[H : K] = 1$, so $H = K$. A kernel is normal, proving the claim.

If $|G| = p^2$, Cauchy's theorem supplies an element of order p . Its cyclic subgroup has order p and index p , so it is normal by the first part.

Technique: a small coset action restricts the order of an image. Compare that bound with the index forced by $K \leq H$. This is the same argument as the lecture's smallest-prime-index theorem.

Foundations: Coset actions, core, Cayley, and smallest-prime index, First Isomorphism Theorem.

4.2.11 — Cycle structure in the regular representation

Dummit–Foote, §4.2, Exercise 11; printed p. 122 (PDF p. 136). Statement normalized from the supplied book; solution independently derived.

Let G be finite and $\pi : G \rightarrow S_G$ be the left regular representation. If $|x| = n$ and $|G| = mn$, prove $\pi(x)$ is a product of m disjoint n -cycles. Deduce that it is odd if and only if n is even and $m = |G|/n$ is odd.

Solution

The permutation $\pi(x)$ sends g to xg . Starting at a point g , its successive images are

$$g, xg, x^2g, \dots, x^{n-1}g,$$

and its next image is $x^n g = g$. If $x^i g = x^j g$, right cancellation gives $x^{i-j} = 1$, so $i \equiv j \pmod{n}$. Thus these n points are distinct and form one cycle of length exactly n . This argument applies to every g , so every cycle has length n .

The cycles partition the mn elements of G , giving exactly m disjoint n -cycles. A cycle of length n is a product of $n - 1$ transpositions, hence has sign $(-1)^{n-1}$. Multiplicativity of sign gives

$$\operatorname{sgn}(\pi(x)) = (-1)^{m(n-1)}.$$

This equals -1 precisely when $m(n - 1)$ is odd, which happens exactly when m is odd and $n - 1$ is odd, or equivalently when m is odd and n is even. For $x = 1$, $n = 1$ and the result says the identity is a product of $|G|$ fixed-point cycles and is even, as expected.

Technique: compute the period of one arbitrary point by cancellation; uniform cycle lengths follow without writing the whole permutation.

Foundations: Coset actions, core, Cayley, and smallest-prime index, First Isomorphism Theorem.

4.2.12 — An odd regular permutation produces an index-2 subgroup

Dummit–Foote, §4.2, Exercise 12; printed p. 122 (PDF p. 136). Statement normalized from the supplied book; solution independently derived.

With G finite and $\pi : G \rightarrow S_G$ its left regular representation, prove that if $\pi(G)$ contains an odd permutation, then G has a subgroup of index 2.

Solution

Compose the regular representation with the sign homomorphism:

$$\chi = \text{sgn} \circ \pi : G \rightarrow \{1, -1\}.$$

A composition of homomorphisms is a homomorphism. The identity maps to 1, and the assumed odd permutation supplies an element mapping to -1 . Thus the image is all of $\{1, -1\}$. Put $H = \ker \chi$. The First Isomorphism Theorem gives

$$G/H \cong \{1, -1\} \cong C_2,$$

so $[G : H] = 2$. In fact H is normal, because it is a kernel.

Technique: compose an available representation with a simpler homomorphism whose image has the desired size. No explicit list of elements of H is needed.

Foundations: Coset actions, core, Cayley, and smallest-prime index, First Isomorphism Theorem.

4.3.13 — Finite groups with two conjugacy classes

Dummit–Foote, §4.3, Exercise 13; printed p. 130 (PDF p. 144). Statement normalized from the supplied book; solution independently derived.

Determine all finite groups having exactly two conjugacy classes.

Solution

The identity always forms a singleton conjugacy class. If there are exactly two classes, all nonidentity elements form one class. Write $n = |G|$; then $n \geq 2$, and for any $x \neq 1$,

$$n - 1 = |\text{Cl}_G(x)| = [G : C_G(x)].$$

A subgroup index divides the order of a finite group, so $n - 1$ divides n . It therefore divides $n - (n - 1) = 1$. Since $n - 1$ is positive, $n - 1 = 1$ and $n = 2$. A group of order 2 is cyclic, hence $G \cong C_2$.

Conversely C_2 is abelian and has two elements, each of which is its own conjugacy class. It has exactly two classes. The trivial group has only one class and is excluded.

Technique: one large orbit produces a divisor only one smaller than the whole group order.

Foundations: Orbit–stabilizer and the kernel of an action, Centralizers, normalizers, and the center.

4.3.19 — Splitting a conjugacy class in a normal subgroup

Dummit–Foote, §4.3, Exercise 19; printed p. 131 (PDF p. 145). Statement normalized from the supplied book; solution independently derived.

Let $H \triangleleft G$, let K be a conjugacy class of G contained in H , and choose $x \in K$. Prove K is a union of $k = [G : HC_G(x)]$ conjugacy classes of H , all of equal size. Deduce an S_n -class of even permutations is either one A_n -class or two A_n -classes of equal size.

For possibly infinite G , interpret k as a cardinal number of classes rather than dividing infinite cardinalities.

Solution

Step 1: partition K by H -conjugacy. Every G -conjugate of x lies in H by normality. Conjugating such an element by H keeps it in K , so K is partitioned into H -conjugacy classes.

Step 2: show these classes have equal size. For $g \in G$, conjugation by g sends the H -class of x onto the H -class of gxg^{-1} :

$$g\{h x h^{-1} : h \in H\}g^{-1} = \{h'(g x g^{-1})h'^{-1} : h' \in H\}.$$

The equality uses $gHg^{-1} = H$. This map is a bijection, with inverse conjugation by g^{-1} . Since every element of K is some gxg^{-1} , all H -classes in K have equal cardinality.

Step 3: count the H -classes using an action. The group G acts transitively on this collection of classes by conjugation. A group element g stabilizes the class of x exactly when $gxg^{-1} = h x h^{-1}$ for some $h \in H$. This equality is equivalent to $h^{-1}g \in C_G(x)$, or $g \in HC_G(x)$. Conversely any $g = hc$ with $c \in C_G(x)$ sends x to $h x h^{-1}$ and stabilizes its H -class. Thus the stabilizer is $HC_G(x)$, a subgroup because H is normal. Orbit–stabilizer gives the number of H -classes as $[G : HC_G(x)]$. This argument also works for infinite groups and avoids cancellation of infinite cardinalities.

Step 4: specialize to A_n . For $n \geq 2$, $A_n \triangleleft S_n$ has index 2. The subgroup $A_n C_{S_n}(x)$ lies between A_n and S_n , so it is either A_n or S_n . Its index is therefore 2 or 1. The previous steps say the S_n -class splits into exactly that many equally sized A_n -classes. For $n = 1$, both groups are trivial and there is one class.

Technique: regard the smaller conjugacy classes themselves as points of an action. This is the normal-subgroup orbit argument from 4.1.9.

Foundations: Orbit–stabilizer and the kernel of an action, Centralizers, normalizers, and the center.

PROBLEM • cu:algebra:midterm-2:section-4-3:pb:df-4-3-23

4.3.23 — Normalizers and conjugates of a maximal subgroup

Dummit–Foote, §4.3, Exercise 23; printed p. 131 (PDF p. 145). Statement normalized from the supplied book; solution independently derived.

Let M be a maximal proper subgroup of G . Prove $N_G(M)$ is either M or G . If G is finite and M is not normal, deduce that at most $(|M| - 1)[G : M]$ nonidentity elements lie in conjugates of M .

Solution

Every element of M normalizes M , so $M \leq N_G(M) \leq G$. Maximality allows only the two possibilities $N_G(M) = M$ or $N_G(M) = G$. The latter means $gMg^{-1} = M$ for every g , which is exactly normality. Thus a nonnormal maximal subgroup is self-normalizing.

Now suppose G finite and M nonnormal. Under the action of G on subgroups by conjugation, the stabilizer of M is $N_G(M) = M$. Orbit–stabilizer therefore says there are $[G : M]$ distinct conjugate subgroups. Each has $|M| - 1$ nonidentity elements because conjugation is a bijection and preserves the identity. The size of a union is at most the sum of the sizes, even when those sets overlap.

Consequently

$$\left| \left(\bigcup_{g \in G} gMg^{-1} \right) \setminus \{1\} \right| \leq (|M| - 1)[G : M].$$

Technique: maximality turns an intermediate subgroup into a dichotomy; orbit–stabilizer then counts the conjugates, and a union bound counts their elements.

Foundations: Orbit–stabilizer and the kernel of an action, Centralizers, normalizers, and the center.

4.3.24 — A finite group is not covered by conjugates of one proper subgroup

Dummit–Foote, §4.3, Exercise 24; printed p. 131 (PDF p. 145). Statement normalized from the supplied book; solution independently derived.

Let $H < G$ with G finite. Prove $G \neq \bigcup_{g \in G} gHg^{-1}$.

Solution

Because G is finite, among the proper subgroups containing H choose one, M , of largest order. Any proper subgroup strictly containing M would have larger order, so M is maximal. Each conjugate of H lies inside the corresponding conjugate of M . It is enough to show the conjugates of M do not cover G .

If $M \triangleleft G$, every conjugate of M is M , and their union is the proper subgroup M .

If M is not normal, put $t = [G : M]$. Since $M < G$, $t \geq 2$. Exercise 4.3.23 bounds the number of nonidentity elements in the union by $(|M| - 1)t$. Adding the common identity gives

$$\left| \bigcup_{g \in G} gMg^{-1} \right| \leq 1 + (|M| - 1)t = |G| - t + 1 < |G|.$$

Therefore this union is proper, and so is the union of conjugates of H .

Technique: enlarge to a maximal subgroup to count conjugates exactly, then exploit the fact that all conjugates share the identity. Finiteness is essential to this counting proof.

Foundations: Orbit–stabilizer and the kernel of an action, Centralizers, normalizers, and the center.

4.4.3 — Bounding automorphisms of D_8

Dummit–Foote, §4.4, Exercise 3; printed p. 137 (PDF p. 151). Statement normalized from the supplied book; solution independently derived.

For the usual generators r, s of D_8 , show that an automorphism has at most two choices for the image of r and at most four choices for the image of s . Deduce $|\text{Aut}(D_8)| \leq 8$.

Solution

An automorphism preserves element orders: if $x^m = 1$, then $\alpha(x)^m = 1$, and applying α^{-1} gives the reverse divisibility of orders. The only elements of order 4 in D_8 are r, r^3 ; the other rotations are $1, r^2$, and each reflection sr^j has order 2. Since r has order 4, $\alpha(r)$ is either r or r^3 .

The five involutions of D_8 are r^2, s, sr, sr^2, sr^3 . Although order preservation initially allows five choices for $\alpha(s)$, it cannot be r^2 . Indeed $\alpha(r)$ already lies in $\langle r \rangle$, and if $\alpha(s) = r^2$ too, the images of the generators would generate a subgroup of $\langle r \rangle$, contradicting surjectivity of α . Thus $\alpha(s)$ must be one of the four reflections.

A homomorphism is determined by its values on generators, because its value on every word is then determined. There are at most $2 \cdot 4 = 8$ possible pairs $(\alpha(r), \alpha(s))$, so at most eight automorphisms. For this upper bound we need not show that every pair actually works.

Technique: use orders to restrict generator images, then use generation/surjectivity to remove choices that preserve orders but cannot yield an automorphism.

Foundations: Automorphisms, inner automorphisms, and characteristic subgroups, Inner automorphisms and characteristic-to-normal transfer, Commutator criterion for abelian quotients.

4.4.8 — Characteristic subgroups — (b)–(c)

Dummit–Foote, §4.4, Exercise 8; printed p. 137 (PDF p. 151). Statement normalized from the supplied book; solution independently derived.

Let $H \leq K \leq G$.

(b) Prove that if H is characteristic in K and K is characteristic in G , then H is characteristic in G . Use this to show the Klein four subgroup of S_4 is characteristic in S_4 .

(c) Give an example where $H \triangleleft K$ and K is characteristic in G but H is not normal in G .

Solution

(b) Transitivity of characteristicity. Let $\alpha \in \text{Aut}(G)$. Since K is characteristic, $\alpha(K) = K$. The restriction $\alpha|_K$ is an automorphism of K : it preserves multiplication, is injective, and maps K onto K . Since H is characteristic in K , $\alpha|_K(H) = H$. Thus $\alpha(H) = H$ for every automorphism of G , proving H characteristic in G .

Apply this to S_4 . Let

$$V = \{1, (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\}.$$

First we verify A_4 is characteristic in S_4 using its commutator description. Because $S_4/A_4 \cong C_2$ is abelian, $S_4' \leq A_4$. For the involutions $a = (1\ 2)$, $b = (2\ 3)$, the commutator is $[a, b] = abab = (ab)^2 = (1\ 3\ 2)$, a 3-cycle. Normality of S_4' then puts every conjugate 3-cycle in S_4' . The 3-cycles generate A_4 : any even permutation is a product of pairs of transpositions, and each pair is either identity, a 3-cycle when the transpositions share one point, or, for disjoint transpositions,

$$(a\ b)(c\ d) = (a\ c\ b)(a\ c\ d).$$

Therefore $S_4' = A_4$. Every automorphism sends commutators to commutators and hence preserves the subgroup they generate, so A_4 is characteristic in S_4 .

Next V is characteristic in A_4 : the elements of order 2 in A_4 are exactly the three double transpositions, so V is the identity together with all order-2 elements. Automorphisms preserve orders, hence preserve V . The transitivity just proved gives V characteristic in S_4 .

(c) Counterexample. Take $G = A_4$, $K = V$, and $H = \langle (1\ 2)(3\ 4) \rangle$. We just proved K characteristic in A_4 . Since K is abelian, every subgroup of K , including H , is normal in K . But

conjugation by $g = (1\ 2\ 3)$ sends the generator of H to

$$g(1\ 2)(3\ 4)g^{-1} = (2\ 3)(1\ 4),$$

which is not in H . Hence H is not normal in G .

Technique: restricting an arbitrary automorphism proves characteristicity. Restricting only conjugations proves normality; these two statements are not interchangeable.

Foundations: Automorphisms, inner automorphisms, and characteristic subgroups, Inner automorphisms and characteristic-to-normal transfer, Commutator criterion for abelian quotients.

4.4.18 — Automorphisms of S_n are inner for $n \neq 6$

Dummit–Foote, §4.4, Exercise 18; printed p. 138 (PDF p. 152). Statement normalized from the supplied book; solution independently derived.

Fix $n \geq 2$ with $n \neq 6$.

- (a) Prove automorphisms of a group permute its conjugacy classes.
- (b) Let K be the class of transpositions in S_n . Show that any other class K' of elements of order 2 has $|K'| \neq |K|$, and deduce automorphisms preserve transpositions.
- (c) Show any automorphism α has

$$\alpha((1\ i)) = (a\ b_i) \quad (2 \leq i \leq n),$$

with $a, b_2, \dots, b_n$ distinct.

- (d) Show the transpositions $(1\ i)$ generate S_n , and conclude $\text{Aut}(S_n) = \text{Inn}(S_n)$.

Solution

(a) Conjugacy classes are mapped onto conjugacy classes. If $\alpha \in \text{Aut}(G)$, then

$$\alpha(gxg^{-1}) = \alpha(g)\alpha(x)\alpha(g)^{-1}.$$

As g runs through G , $\alpha(g)$ also runs through G because α is surjective. Therefore $\alpha(\text{Cl}_G(x)) = \text{Cl}_G(\alpha(x))$. The restriction of α is a bijection of these classes, so their sizes agree.

(b) Compare the involution classes by counting. An involution in S_n is a product of r disjoint transpositions, with $1 \leq r \leq \lfloor n/2 \rfloor$. The number with this cycle type is

$$c_r = \frac{n!}{2^r r! (n - 2r)!}.$$

To derive it, order $2r$ distinct points in $n!/(n - 2r)!$ ways, pair successive points, and divide by 2^r for the order within each pair and by $r!$ for the order of the pairs. The transposition class has $c_1 = n(n - 1)/2$ elements. Equality $c_r = c_1$ for $r \geq 2$ is equivalent to

$$\frac{(n - 2)!}{(n - 2r)!} = 2^{r-1} r!.$$

For $r = 2$, this would mean $(n - 2)(n - 3) = 4$. At $n = 4$ the left side is 2 and at $n = 5$ it is 6; it increases thereafter, so equality never occurs. For $r = 3$, the smallest permitted n is 6, where the left side is $4! = 24 = 2^2 \cdot 3!$. It strictly increases for $n > 6$, so equality occurs only at $n = 6$. For $r \geq 4$, at the smallest permitted $n = 2r$ the ratio of the left side to the right side is

$$R_r = \frac{(2r - 2)!}{2^{r-1}r!}.$$

Here $R_4 = 720/192 > 1$, and

$$\frac{R_{r+1}}{R_r} = \frac{(2r)(2r - 1)}{2(r + 1)} > 1 \quad (r \geq 4).$$

Thus $R_r > 1$ for all $r \geq 4$; increasing n only increases the left side. There are no more equalities. Since $n \neq 6$, the transposition class is the unique involution class of its size. Part (a) and order preservation force every automorphism to send transpositions to transpositions.

(c) Show the images form a star. Put $t_i = \alpha((1 \ i))$. They are distinct transpositions because α is injective. For $i \neq j$, $(1 \ i)(1 \ j)$ has order 3, so $t_i t_j$ has order 3. Two distinct transpositions have product of order 3 exactly when their two-point supports intersect in one point: disjoint ones commute and their product has order 2, whereas $(a \ b)(a \ c)$ is a 3-cycle. Thus the supports of the t_i intersect pairwise.

For $n = 2$, there is only one image transposition; choose its two points as a, b_2 . For $n = 3$, the two image transpositions share a point, which can be named a , with their other points b_2, b_3 .

For $n \geq 4$, take two images with supports $\{a, b\}$ and $\{a, c\}$. Any further two-point support meeting both either contains a or is $\{b, c\}$. If $\{b, c\}$ occurs, every support meeting all three must be one of $\{a, b\}, \{a, c\}, \{b, c\}$: a support using a new point misses at least one of these three. Hence a pairwise intersecting family with no common point has at most three members, forming a triangle. For $n \geq 5$ our family has $n - 1 \geq 4$ members, so a triangle is impossible.

The remaining case is $n = 4$. The three original transpositions $(1 \ 2), (1 \ 3), (1 \ 4)$ have product $(1 \ 4 \ 3 \ 2)$ of order 4. If their images formed a triangle on three points, their product would act on only those three points and be odd, so it would be a transposition and have order 2; the odd elements of S_3 are exactly its transpositions. This contradicts order preservation of that product. Thus here too all images share a point a . Write them as $(a \ b_i)$. Distinctness implies the b_i are distinct and none equals a , so together these are all n points.

(d) Identify the automorphism. For distinct $i, j \neq 1$,

$$(i \ j) = (1 \ i)(1 \ j)(1 \ i),$$

so the star transpositions generate all transpositions and hence all S_n . Define $\tau \in S_n$ by $\tau(1) = a$, $\tau(i) = b_i$. Part (c) makes this a bijection of the n points. Conjugation by τ agrees with α on every star generator:

$$\tau(1\ i)\tau^{-1} = (a\ b_i) = \alpha((1\ i)).$$

Two homomorphisms agreeing on generators agree on every word, so $\alpha(g) = \tau g \tau^{-1}$ for all $g \in S_n$. Thus every automorphism is inner; every inner automorphism is already an automorphism, proving equality.

Technique: invariant class sizes identify a generating family; elementary support intersections then recover a relabeling of the points. The $n = 4$ triangle and the $n = 6$ class-size coincidence require separate checks.

Foundations: Automorphisms, inner automorphisms, and characteristic subgroups, Inner automorphisms and characteristic-to-normal transfer, Commutator criterion for abelian quotients.

4.4.20 — The Thompson subgroup — (a)–(b)

Dummit–Foote, §4.4, Exercise 20; printed p. 139 (PDF p. 153). Statement normalized from the supplied book; solution independently derived.

For a finite group P , let $d(P)$ be its minimum number of generators, and let

$$m(P) = \max\{d(A) : A \leq P \text{ abelian}\},$$

$$J(P) = \langle A : A \leq P \text{ abelian and } d(A) = m(P) \rangle.$$

Take $d(1) = 0$.

(a) Prove $J(P)$ characteristic in P .

(b) List every abelian subgroup A attaining $d(A) = m(P)$ for $P = Q_8, D_8, D_{16}, QD_{16}$, where

$$QD_{16} = \langle \sigma, \tau \mid \sigma^8 = \tau^2 = 1, \sigma\tau = \tau\sigma^3 \rangle.$$

Only parts (a) and (b) are assigned.

Solution

(a) Automorphisms preserve the defining family. Let $\alpha \in \text{Aut}(P)$ and let $A \leq P$ be abelian. Its image $\alpha(A)$ is a subgroup and is abelian, since images of commuting elements commute. An isomorphism preserves the minimum number of generators: the images of a generating set generate the image, giving $d(\alpha(A)) \leq d(A)$, and applying α^{-1} gives the reverse inequality. Thus $d(\alpha(A)) = d(A)$. Consequently α permutes exactly the abelian subgroups attaining $m(P)$. The subgroup generated by this family is therefore preserved: $\alpha(J(P)) = J(P)$. This is characteristicity.

(b) Q_8 . The proper nontrivial subgroups are $\langle -1 \rangle, \langle i \rangle, \langle j \rangle, \langle k \rangle$, and each is cyclic. The whole group is nonabelian. Thus $m(Q_8) = 1$, and all four listed subgroups attain it. In particular, the order-2 subgroup must be included, not just the maximal cyclic subgroups.

D_8 . The two subgroups

$$\langle r^2, s \rangle, \quad \langle r^2, sr \rangle$$

are Klein four groups and require two generators. The rotation subgroup and all smaller cyclic subgroups require at most one, and D_8 itself is nonabelian. Its subgroup lattice therefore shows $m(D_8) = 2$ and these are exactly the attaining abelian subgroups.

D16. Use $|r| = 8$, $s^2 = 1$, $srs = r^{-1}$. An abelian subgroup contained in $\langle r \rangle$ is cyclic. If an abelian subgroup contains a reflection sr^j , any rotation r^k in it must satisfy

$$sr^j r^k = r^k sr^j,$$

which is equivalent to $r^k = r^{-k}$, so $k \equiv 0$ or $4 \pmod{8}$. Two reflections sr^i, sr^j commute exactly when $2(i - j) \equiv 0 \pmod{8}$, so their indices differ by 0 or 4. Hence any noncyclic abelian subgroup with a reflection is exactly one of

$$\langle r^4, sr^j \rangle \quad (j = 0, 1, 2, 3).$$

Each is V_4 and needs two generators. These four, and no others, attain $m(D_{16}) = 2$.

QD16. The relation gives $\tau\sigma\tau = \sigma^3$. Every element outside $\langle \sigma \rangle$ is $\tau\sigma^i$, and

$$(\tau\sigma^i)^2 = \sigma^{4i}.$$

If i is even this is an involution; if i is odd it has order 4 and its square is σ^4 . A rotation σ^k commutes with $\tau\sigma^i$ exactly when $2k \equiv 0 \pmod{8}$, so $k \equiv 0, 4$. For two outside elements, their products have exponents $3i + j$ and $3j + i$, respectively. They commute exactly when $i - j \equiv 0 \pmod{4}$.

It follows that an abelian subgroup containing an outside element lies in $\langle \sigma^4, \tau\sigma^i \rangle$: its rotations are $1, \sigma^4$, and its other outside elements can only be $\tau\sigma^i$ and $\tau\sigma^{i+4}$. For odd i this group is cyclic of order 4, because $(\tau\sigma^i)^2 = \sigma^4$. For even i it is V_4 . Thus the only abelian subgroups needing two generators are

$$\langle \sigma^4, \tau \rangle, \quad \langle \sigma^4, \tau\sigma^2 \rangle,$$

and $m(QD_{16}) = 2$.

Technique: an invariant definition gives characteristicity. For the explicit lists, separate cyclic rotation subgroups from subgroups containing an outside element, then calculate exactly when generators commute.

Foundations: Automorphisms, inner automorphisms, and characteristic subgroups, Inner automorphisms and characteristic-to-normal transfer, Commutator criterion for abelian quotients.

4.6.4 — Three-cycles generate A_n

Dummit–Foote, §4.6, Exercise 4; printed p. 151 (PDF p. 165). Statement normalized from the supplied book; solution independently derived.

Prove A_n is generated by all 3-cycles for every $n \geq 3$.

Solution

Each 3-cycle is a product of two transpositions, so it is even. Hence the subgroup L generated by all 3-cycles satisfies $L \leq A_n$.

For the reverse inclusion, take $\sigma \in A_n$. Since transpositions generate S_n , write σ as a product of transpositions. Evenness says that the number of factors is even, so group them in consecutive pairs. It is enough to express each pair as a product of 3-cycles.

There are three possibilities:

1. If the transpositions are equal, their product is the identity and contributes no 3-cycles.
2. If they share exactly one point, after naming that point a the pair is $(a\ b)(a\ c) = (a\ c\ b)$, a 3-cycle.
3. If they are disjoint, then for four distinct points,

$$(a\ b)(c\ d) = (a\ c\ b)(a\ c\ d).$$

To verify the last identity, apply the right side from right to left: it sends $a \mapsto b$, $b \mapsto a$, $c \mapsto d$, and $d \mapsto c$, and fixes all other points.

Thus every pair belongs to L , and their product σ does too. Therefore $A_n \leq L$, proving $A_n = L$. When $n = 3$ the disjoint case simply cannot occur, and the same argument still works.

Technique: pair transpositions because the sign forces an even count. This generation theorem does not require the much stronger simplicity theorem.

Foundations: Technique — Conjugate supports and pair transpositions.

5.1.7 — Permuting direct-product factors

Dummit–Foote, §5.1, Exercise 7; printed p. 156 (PDF p. 170). Statement normalized from the supplied book; solution independently derived.

Let $G_1, \dots, G_n$ be groups and $\pi \in S_n$. Prove

$$\varphi_\pi : G_1 \times \cdots \times G_n \longrightarrow G_{\pi^{-1}(1)} \times \cdots \times G_{\pi^{-1}(n)},$$

$$\varphi_\pi(g_1, \dots, g_n) = (g_{\pi^{-1}(1)}, \dots, g_{\pi^{-1}(n)})$$

is an isomorphism.

Solution

Step 1: check the codomain. For each position j , the output coordinate is $g_{\pi^{-1}(j)} \in G_{\pi^{-1}(j)}$. It therefore belongs to the prescribed factor in that position, making the map well defined even when the groups differ.

Step 2: preserve multiplication. Products in a direct product are coordinatewise. For tuples $g = (g_i)$ and $h = (h_i)$, the j th coordinate of $\varphi_\pi(gh)$ is $g_{\pi^{-1}(j)}h_{\pi^{-1}(j)}$. This is also the j th coordinate of $\varphi_\pi(g)\varphi_\pi(h)$. Equality in every coordinate proves the homomorphism property.

Step 3: construct the inverse. For a target tuple $k = (k_1, \dots, k_n)$, define

$$\psi(k) = (k_{\pi(1)}, \dots, k_{\pi(n)}).$$

The i th coordinate lies in G_i , since $k_{\pi(i)} \in G_{\pi^{-1}(\pi(i))} = G_i$. Then

$$(\psi\varphi_\pi(g))_i = g_{\pi^{-1}(\pi(i))} = g_i,$$

and

$$(\varphi_\pi\psi(k))_j = k_{\pi(\pi^{-1}(j))} = k_j.$$

Thus ψ is a two-sided inverse, so φ_π is bijective and hence an isomorphism.

Technique: check a tuple map one coordinate at a time. The inverse indices ensure that the map goes to the stated reordered product.

Foundations: Direct products, complements, and extremal subgroups.

PROBLEM • cu:algebra:midterm-2:section-5-4:pb:df-5-4-15

5.4.15 — Two abelian quotients give an abelian intersection quotient

Dummit–Foote, §5.4, Exercise 15; printed p. 174 (PDF p. 188). Statement normalized from the supplied book; solution independently derived.

If $A, B \triangleleft G$ and both G/A and G/B are abelian, prove $G/(A \cap B)$ is abelian.

Solution

First $A \cap B \triangleleft G$, because conjugation preserves each of A and B and hence their intersection.

For arbitrary $x, y \in G$, commutativity in G/A gives

$$x^{-1}y^{-1}xy \in A.$$

The same argument in G/B gives $x^{-1}y^{-1}xy \in B$. Therefore every commutator lies in $A \cap B$. In the quotient by this intersection, for arbitrary cosets $\bar{x}, \bar{y}$, this says

$$\bar{x}^{-1}\bar{y}^{-1}\bar{x}\bar{y} = 1,$$

which is equivalent to $\bar{x}\bar{y} = \bar{y}\bar{x}$. All quotient elements commute, so the quotient is abelian.

Equivalent concise view. The commutator criterion says $G' \leq A$ and $G' \leq B$, hence $G' \leq A \cap B$. The same criterion then gives the conclusion. The element calculation above explains every step of that shorthand.

Technique: express both assumptions as membership of the same element, then intersect. There is no need to assume $G = AB$ or use a direct-product isomorphism.

Foundations: Commutator criterion for abelian quotients.

5.4.19 — Perfect subgroups and the maximal perfect subgroup

Dummit–Foote, §5.4, Exercise 19; printed p. 174 (PDF p. 188). Statement normalized from the supplied book; solution independently derived.

A group H is perfect if $H' = H$.

- (a) Prove every nonabelian simple group is perfect.
- (b) If H, K are perfect subgroups of G , prove $\langle H, K \rangle$ perfect. Extend this to a subgroup generated by any collection of perfect subgroups.
- (c) Prove every conjugate of a perfect subgroup is perfect.
- (d) Prove every group has a unique maximal perfect subgroup and that subgroup is normal.

Solution

(a) Simplicity forces the commutator subgroup to be everything. The commutator subgroup H' is normal in H , so simplicity gives $H' = 1$ or $H' = H$. If $H' = 1$, every commutator is the identity, so every pair of elements commutes and H is abelian. This contradicts the nonabelian hypothesis. Therefore $H' = H$.

(b) Generation preserves perfectness. Let $L = \langle H, K \rangle$. Commutators of elements of H are also commutators of elements of L , so $H' \leq L'$. Since $H' = H$, we have $H \leq L'$. Likewise $K \leq L'$. A subgroup containing both H and K contains the subgroup they generate, so $L \leq L'$. The reverse inclusion $L' \leq L$ is automatic, since commutators of elements of L are elements of L . Thus $L' = L$.

For a family $(H_i)_{i \in I}$ of perfect subgroups, the same reasoning gives $H_i = H'_i \leq L'$ for every i , where $L = \langle H_i : i \in I \rangle$. Thus $L \leq L'$ and equality follows. The empty family generates 1 , which is also perfect, so no exception is needed.

(c) Conjugation carries derived subgroups to derived subgroups. For $g \in G$,

$$[gh_1g^{-1}, gh_2g^{-1}] = g[h_1, h_2]g^{-1}.$$

Taking the subgroups generated by both sides gives $(gHg^{-1})' = gH'g^{-1}$. If $H' = H$, this becomes $(gHg^{-1})' = gHg^{-1}$, proving the conjugate perfect.

(d) Generate from all perfect subgroups. Let P be the subgroup generated by every perfect subgroup of G . This is a set of subgroups, and it includes 1 . Part (b) shows P itself is perfect. By construction every perfect subgroup is contained in P . Thus P is a greatest perfect subgroup, a stronger statement than maximality. If Q were any maximal perfect subgroup, then $Q \leq P$ and P is perfect, so maximality would force $Q = P$. This proves uniqueness.

For $g \in G$, part (c) says conjugation permutes all perfect subgroups. Consequently it preserves their generated subgroup, so $gPg^{-1} = P$. Hence $P \triangleleft G$. Indeed every automorphism preserves perfectness by the same commutator calculation, so P is characteristic as well.

Technique: commutator subgroups are monotone under subgroup inclusion. Generating all subgroups with a property can provide a greatest subgroup, provided that property survives generation.

Foundations: Commutator criterion for abelian quotients.

Lecture map and scope boundaries

Source map: supplied Week 4, Week 5, and Week 6 lecture PDFs, all 36 pages visually inspected. Page numbers above are PDF pages.

The supplied homework list determines this study's exercise scope. The three lecture PDFs were visually inspected; they explain the preferred proof methods, rather than adding more homework.

Week 4 (9 PDF pages). Pages 1–3 develop subgroup products, the criterion $HK = KH$, orbit-stabilizer, and the product-order formula. Page 4 counts subsets under a symmetric-group action. Pages 5–8 prove the first, second, third, and lattice isomorphism theorems. Page 9 introduces simple groups and explicitly says the Feit–Thompson result is usually off limits. No solution here uses it.

Week 5 (13 PDF pages). Pages 1–3 cover composition series and Jordan–Hölder. Pages 4–5 cover solvability, commutator subgroups, the derived sequence, and extensions. Pages 6–9 treat symmetric groups, sign, and alternating groups. Page 10 proves normality for smallest-prime index by a coset action. Pages 11–13 develop conjugacy, cycle type, and integer partitions.

Week 6 (14 PDF pages). Pages 1–4 treat class splitting and simplicity of A_n . Page 5 gives the class equation and p -group center argument. Pages 6–7 prove orbit counting by fixed points and the fixed-point-free consequence. Pages 8–10 cover automorphisms, inner automorphisms, and characteristic subgroups. Pages 11–12 cover external and internal direct products. Pages 13–14 state the finite abelian-group structure theorem and begin semidirect products.

Book-only details required by the homework. The explicit lattice examples, block/primitivity problem, Thompson-subgroup definition, and maximal perfect subgroup are developed in the assigned book sections or exercises, and their solutions use the elementary methods above. The subgroup P in 3.3.9 is given; full Sylow theory is not used to manufacture it.

Notation. The textbook commutator is $x^{-1}y^{-1}xy$. The lecture's commutator generating family gives the same subgroup, but individual expressions differ. Pointwise and setwise stabilizers are distinguished in the definitions. D_{2n} has order $2n$; C_m denotes a cyclic group of order m , written Z_m in the textbook.

Boundary correction. Exercise 4.1.7(d)'s maximal-stabilizer equivalence needs more than one point. The solution explains the singleton exception rather than silently asserting the unqualified statement.

The later finite-abelian-group classification and semidirect-product construction are identified here as lecture context; this study does not assert they are extra exam requirements.